Highlights

HDQE: Hybrid Deep Quantum Embeddings with Pauli Readouts and Learnable Fusion

- Proposes a Hybrid Deep Quantum Embedding (HDQE) framework combining MARBERT with quantum feature mapping.
- Explores 134 configurations across qubit count (Q), depth (L), fusion, and entanglement patterns.
- Achieves 0.496 ± 0.020 macro-F1, outperforming classical MARBERT by $+$**3.8 points** ($p < 0.001$).
- Demonstrates that *weighted fusion* and shallow quantum circuits ($L = 2$) provide optimal trade-offs.
- Establishes a scalable, hardware-efficient benchmark for future quantum-enhanced NLP models.

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HDQE: Hybrid Deep Quantum Embeddings with Pauli Readouts and Learnable Fusion

Abstract

This paper introduces Hybrid Deep Quantum Embedding (HDQE), a quantum-inspired representation learning framework that integrates pre-trained Arabic Transformer embeddings with parameterized quantum circuits. HDQE combines classical contextual embeddings with quantum circuit design choices including qubit count (Q), circuit depth (L), entanglement topology, Pauli-based measurement readouts, and learnable fusion to construct compact hybrid embeddings. We systematically evaluate 134 HDQE variants built on MARBERT, AraBERT, and XLM-R, and compare them against transformer baselines and parameter-matched classical control models. Across all settings, HDQE consistently improves macro-F1, with the best configuration achieving 0.496 ± 0.020, exceeding MARBERT by $+3.8$ points with statistical significance ($p < 0.001$). Our results show that shallow circuits ($L \leq 3$) and weighted fusion provide the most stable expressivity–generalization trade-off. All quantum components are classically simulated, and no claims of quantum hardware advantage are made. HDQE provides a reproducible benchmark for studying quantum-inspired feature mappings in Arabic NLP.

Keywords: MARBERT, Quantum embedding, Quantum NLP, Arabic sentiment analysis, Entanglement, Weighted fusion

1. Introduction

Natural Language Processing (NLP) is considered a core and fundamental technology that drives modern information extraction, intelligent search, and human–computer interaction. A key challenge in NLP is achieving embedding generalization the ability of a learned representation to preserve semantic and contextual meaning across domains, tasks, and dialects. The Arabic language is distinguished by its high morphological complexity, diglossia, and regional diversity, making this challenge particularly formidable. Further, the Arabic language is known for the diversity of its dialects with each dialect introduces unique phonological, orthographic, and syntactic variations, resulting in sparse distributions and semantic drift in vector space models.

Transformer architectures form the backbone of modern NLP, such as BERT, AraBERT [1], and multilingual XLM-R [2]. These models provide strong contextual understanding. However, they reamin confiend to real-valued manifolds, limiting their ability to capture non-linear semantic interactions and long-range dependencies. This limitation motivates exploration of richer representational spaces capable of capturing the probabilistic, multi-state nature of linguistic meaning.

Quantum Machine Learning (QML) is a new and rich area for research and exploration. The latest research in this area has introduced new possibilities for improving embedding methods. In these models, information is expressed through quantum states in complex Hilbert spaces, with both amplitudes and phases carrying representational value. Parameterized quantum circuits (PQCs) apply unitary transformations and entanglement operations that naturally introduce non-linear interactions beyond classical kernels [3, 4]. This process produces compact yet highly expressive embeddings suitable for tasks requiring fine-grained semantic discrimination.

Even with these theoretical advantages in place, the integration of quantum feature mapping with large pre-trained language models remains underexplored, especially in languages with complex morphology, such

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as Arabic. Existing Quantum NLP (QNLP) works largely focus on small English corpora or toy datasets and rarely examine hyperparameter effects such as (Q), (L), entanglement topology, or hybrid fusion strategies. To address this gap, this research proposes the Hybrid Deep Quantum Embedding (HDQE) framework a scalable architecture coupling Arabic transformer encoders (e.g., MARBERT, AraBERT) with configurable PQCs for enhanced embedding generalization.

The principal contributions of this research are as follows:

- The introduction of HDQE architecture, which fuses classical transformer embeddings with parameterized quantum circuits, systematically varying (Q), (L), and entanglement structure.
- The design of two classical control families CL3Q (random Fourier projection) and CLMLP (autoencoder bottleneck) to isolate quantum contributions from dimensional effects.
- Performing a large-scale empirical evaluation over 134 model configurations, creating the first Arabic quantum embedding benchmark and reporting statistically significant improvements over MARBERT using five seed cross validation and bootstrap tests.
- Providing an empirical and theoretical evidence that quantum-enhanced embeddings can improve representational diversity and generalization while remaining feasible on near-term (NISQ) quantum hardware.

This study reframes Arabic sentiment analysis as a proxy task to evaluate embedding quality, using the ASTD dataset for empirical grounding but focusing on generalizable embedding design applicable to broader NLP domains.

The remainder of this paper is organized as follows. Section 2 reviews the linguistic and technical foundations of Arabic embeddings and quantum representation learning. Section 3 surveys related work. Section 4 details the proposed methodology. Section 5 describes the experimental setup while Section 6 describes the experimental results. Geometric Diagnostics of Embedding Spaces on the Unit Sphere is covered in Section 7. Section 8 presents a computational complexity and scalability analysis, followed by Section 9, which interprets the observed improvements and discusses implications for future quantum NLP research. Finally, Section 10 provides a conclusion.

2. Background

This section provides the conceptual and technical background necessary to contextualize the proposed HDQE framework. We first review advances in Arabic text representation and transformer-based contextual embeddings, which form the classical foundation of our approach. We then introduce the core principles of vector space representations and similarity measures that underlie standard NLP pipelines. Building on this classical foundation, we outline the basics of quantum states, parameterized quantum circuits, and quantum feature encoding, highlighting how quantum-inspired representations can enrich embedding geometry.

2.1. Arabic Text Representation and Contextual Embeddings

The representation of Arabic text has evolved substantially with the adoption of deep contextual language models, particularly transformer-based architectures. Earlier approaches based on surface-form features or static word embeddings struggled to capture the linguistic properties of Arabic, including nonconcatenative morphology, flexible word order, and widespread dialectal variation. These properties directly affect how semantic information is distributed across text and therefore place strong requirements on the underlying embedding representations [5, 6].

The building blocks of Arabic morphology are the root-pattern. The core idea hinges on the concept that consonantal roots and their corresponding vocalic and affixal patterns interact in distinct ways to create meaning. This gives roots the ability to spawn multiple meanings depending on how the root and patterns

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interact to form a particular word. So even synonomous roots can create a high degree of semantic syntactic relations. In addition, Arabic exhibits a high degree of Diglossia, in which Modern Standard Arabic is mixed with and code-switched into Colloquial Arabic in fast speech and informal writing. This has the effect of increasing potential ambiguity in a given context [7, 8]. This means that in such contexts, a representation model must flexibly and dynamically encode meaning, rather than rigidly relying on a particular lexical identity.

Contextual embeddings solve this problem. They create embeddings based on the entire context of the sentence in which the token appears. This means that each token embedding is, in a sense, a function of the entire input sequence, since it is computed via a self-attention mechanism. Given an input sequence of tokens, the model operates on the ordered set

$$(x_1, x_2, \ldots, x_T), \tag{1}$$

where x_t denotes the t-th token in the sequence and T is the sequence length. The transformer encoder maps this sequence to a corresponding sequence of hidden states

$$(h_1, h_2, \ldots, h_T), \tag{2}$$

where each $h_t \in \mathbb{R}^d$ is a d-dimensional contextual embedding associated with token x_t. Here, d denotes the embedding dimensionality determined by the model architecture. Each hidden state h_t integrates information from all other tokens in the sequence, allowing the representation to reflect both local and long-range dependencies.

The interpretations of Arabic words vary depending on several modifiers that are not part of the words, such as negation markers, sentence markers, or sentence changes. Arabic-specific transformer models have outperformed multilingual models in producing contextual embeddings across all varieties of Arabic and across all research areas, for tasks that require sentiment and attitude identification [5]. Thus, the Arabic contextual embeddings mitigate the adverse effects of Arabic morphology, orthography, and sparsity.

For sentiment classification, token-level embeddings are commonly aggregated or combined into a static, less heterogeneous dimension that characterizes all words in each sentence. A common approach is mean pooling, defined as

$$\mu = \frac{1}{T} \sum_{t=1}^{T} h_t, \tag{3}$$

where $\mu \in \mathbb{R}^d$ denotes the sentence-level embedding, T is the number of tokens in the sequence, and h_t is the contextual embedding of the t-th token. An alternative strategy is to select a designated classification embedding, expressed as

$$\mu = h_{[\mathrm{CLS}]}, \tag{4}$$

where $h_{[\mathrm{CLS}]}$ represents the hidden state associated with a special classification token inserted at the beginning of the sequence. Both strategies aim to produce a compact semantic summary suitable for downstream classifiers.

In the context of Arabic sentiment analysis, the role of aggregation functions becomes more critical, given that sentiment cues are more often spread across many tokens as opposed to being concentrated in a lexeme. Although contextualized embeddings have become standard, they are still represented as real-valued vectors, with similarity typically computed as the inner product or cosine distance. These distance metrics assume that relationships between a set of vectors are semantic and can be modeled with linear, and in some cases mildly nonlinear, models in $\mathbb{R}^d$. The most contemporary embeddings have, to some extent, contextualized Arabic sentiment, clustering it with, as some articles suggest, pragmatically irrelevant classes, indicating a degree of embedding geometry similar to that used in conventional approaches [9, 10].

From a methodological point of view, contextual embeddings provide a robust, validated foundation for Arabic text modeling. They encode rich syntactic and semantic information while exposing a fixed-dimensional interface that downstream learning components can further transform. A precise understanding

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of how these embeddings are constructed, normalized, and aggregated is therefore essential for methods that seek to extend or enrich Arabic representations, forming the representational baseline for the methodology introduced in subsequent sections.

2.2. Vector Space Representations and Similarity Measures

Currently, in most text processing and NLP tasks, text is first converted into numeric (real-valued) vectors. After language models generate contextual embeddings, these embeddings lie in a finite-dimensional vector space (embedding space), and the distances between points are assumed to encode the meanings of the texts they represent. This vector space is a formalism that is particularly helpful for working with a range of machine learning models, whether linear, kernel, or neural networks [11].

Formally, a text instance (e.g., a sentence or document) is mapped to a vector

$$\mathbf{x} \in \mathbb{R}^d, \tag{5}$$

where d denotes the embedding dimensionality and $\mathbb{R}^d$ is a real-valued Euclidean space. The vector $\mathbf{x}$ typically results from pooling or selecting token-level contextual embeddings, as described in the previous subsection. In practice, the absolute scale of $\mathbf{x}$ may vary across instances due to differences in token length, lexical composition, or model dynamics, which motivates the use of normalization.

A common preprocessing step is ℓ_2 normalization, defined as

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}, \tag{6}$$

where $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^{d} x_i^2}$ is the Euclidean norm of the vector $\mathbf{x}$ and $\hat{\mathbf{x}}$ denotes the normalized representation. Normalization projects all embeddings onto the unit hypersphere, reducing sensitivity to magnitude differences and emphasizing directional information. This step is crucial when computing similarity using inner products or cosine-based measures.

Similarity between two text representations $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ is often quantified using the inner product

$$s(\hat{\mathbf{x}}, \hat{\mathbf{y}}) = \hat{\mathbf{x}}^\top \hat{\mathbf{y}}, \tag{7}$$

where $\hat{\mathbf{x}}^\top \hat{\mathbf{y}}$ denotes the dot product between the two normalized vectors. When vectors are ℓ_2-normalized, this quantity is equivalent to cosine similarity and takes values in the interval $[-1, 1]$. High similarity indicates that the vectors point in similar directions in the embedding space, which is interpreted as semantic relatedness [12, 13].

These geometric assumptions directly affect classification. For grief downstream tasks, we use a variety of linear classifiers. This includes, but is not limited to, logistic regression and support vector machines, which implicitly slice the embedding space via hyperplanes, with the orientation of the hyperplanes determined by inner-product relationships. Therefore, while the real-valued vector space is efficient and interpretable, it constrains the problem to the architecture of the embedding space. With respect to inner-product similarity, the projection of representations on a hyperplane and the smoothness of the angular distance are assumed to be representative of the relationship between including distance the projections represent on the hyperplane. This means that the space in which these configurations exhibit high semantic overlap is left unbounded, as it is not representable. Consequently, the underlying hyperplanes cannot be used to derive a linear classification decision. Hence, the geometric properties of the problem space must be understood to use methods that either transform or augment the embedding representations to improve classification.

2.3. Quantum States and Parameterized Quantum Circuits

Quantum computation is formulated within the mathematical framework of complex Hilbert spaces, where information is represented using quantum states rather than classical vectors. The fundamental unit

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of quantum information is the *qubit*, which generalizes the classical binary variable [14]. A state vector describes a single qubit as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \tag{8}$$

where $|0\rangle$ and $|1\rangle$ denote the computational basis states, and $\alpha, \beta \in \mathbb{C}$ are complex-valued amplitudes satisfying the normalization constraint

$$|\alpha|^2 + |\beta|^2 = 1. \tag{9}$$

Here, $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of observing the qubit in states $|0\rangle$ and $|1\rangle$, respectively, upon measurement.

For a system of Q qubits, the joint quantum state resides in a 2^Q-dimensional Hilbert space, and can be expressed as

$$|\psi\rangle = \sum_{k=0}^{2^Q-1} c_k|k\rangle, \tag{10}$$

where $|k\rangle$ denotes a computational basis state indexed by the binary representation of k, $c_k \in \mathbb{C}$ are complex amplitudes, and $\sum_k |c_k|^2 = 1$. This exponential growth of the state space underlies the expressive capacity of multi-qubit systems.

Quantum states evolve through unitary transformations. In practical quantum algorithms, these transformations are implemented using *quantum circuits*, which consist of sequences of quantum gates applied to qubits [15]. A parameterized quantum circuit (PQC) is a circuit whose gates depend on a set of real-valued parameters [14]. Formally, a PQC defines a unitary operator

$$U(\boldsymbol{\theta}) = U_L(\theta_L) \cdots U_2(\theta_2) U_1(\theta_1), \tag{11}$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_L)$ denotes the set of circuit parameters, L is the number of layers, and each $U_\ell(\theta_\ell)$ represents a parameterized unitary transformation composed of a single-qubit rotations and, optionally, multi-qubit entangling gates.

Single-qubit rotation gates, such as rotations around the Y or Z axes of the Bloch spheres are commonly used and are defined as

$$R_Y(\theta) = \exp\left(-i\frac{\theta}{2}Y\right), \tag{12}$$

where $\theta \in \mathbb{R}$ is a rotation angle and Y is the Pauli-Y operator. Entangling gates, such as controlled-NOT (CNOT), introduce correlations between qubits and cannot be decomposed into independent single-qubit operations [16]. Together, these components allow PQCs to generate a wide range of quantum states from an initial reference state, typically $|0\rangle^{\otimes Q}$.

2.4. Quantum Feature Encoding and Measurement

To integrate quantum circuits into classical machine learning pipelines, classical data must first be encoded into quantum states. This process, known as quantum feature encoding, maps a real-valued input vector to the parameters of a quantum circuit [17]. Let

$$\mathbf{x} = (x_1, x_2, \ldots, x_d) \in \mathbb{R}^d \tag{13}$$

denote a classical feature vector of dimension d. A common encoding strategy associates components of $\mathbf{x}$ with rotation angles in a parameterized encoding unitary

$$U_{\text{enc}}(\mathbf{x}) = \prod_{i=1}^{Q} R_Y(\omega_i x_i), \tag{14}$$

where Q is the number of qubits, $\omega_i \in \mathbb{R}$ are scaling factors, and x_i denotes the i-th component of the input vector (or a projected version thereof). Applying this unitary to the reference state produces an

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input-dependent quantum state

$$|\psi(\mathbf{x})\rangle = U_{\text{enc}}(\mathbf{x})|0\rangle^{\otimes Q}. \tag{15}$$

This state serves as the quantum representation of the classical data.

After encoding and subsequent circuit operations, information must be extracted from the quantum system through measurement. Measurements are performed with respect to quantum observables, which are Hermitian operators [18]. A commonly used observable is the Pauli-Z operator acting on a given qubit. The expectation value of this observable for qubit i is given by:

$$z_i = \langle\psi(\mathbf{x})|Z_i|\psi(\mathbf{x})\rangle, \tag{16}$$

where Z_i denotes the Pauli-Z operator applied to the i-th qubit and identity operators on all others. The scalar $z_i \in [-1, 1]$ represents the difference between the probabilities of measuring the states $|0\rangle$ and $|1\rangle$ on that qubit.

By measuring multiple qubits, one obtains a real-valued feature vector

$$\mathbf{z} = (z_1, z_2, \ldots, z_Q) \in \mathbb{R}^Q, \tag{17}$$

where each component corresponds to an expectation value of a chosen observable. These measurement outcomes provide a deterministic, real-valued interface between quantum circuits and classical learning algorithms. As such, the quantum feature encoding followed by measurement yields representations that can be directly combined with classical features or supplied to standard classifiers, enabling hybrid quantum–classical processing pipelines.

3. Related Work

This section reviews prior work relevant to the proposed HDQE framework, spanning three complementary research directions. We first summarize advances in Arabic NLP and transformer-based architectures, which establish the state of the art in contextual representation learning for morphologically rich languages. We then review key developments in quantum natural language processing and parameterized quantum circuits, highlighting how quantum-inspired representations have been explored for semantic modeling. Finally, we discuss hybrid quantum - classical models that integrate pretrained neural encoders with quantum feature mappings.

3.1. Arabic NLP and Transformer Advances

Arabic NLP has progressed from rule-based and lexicon-driven sentiment scoring approaches [19, 20] toward advanced contextualized models built on transformer architectures. Early embedding research, including AraVec [21] and FastText-Arabic [22], introduced static word-level representations that enhanced baseline sentiment and topic classification, however, struggled with polysemy and contextual variability. Following this, AraBERT [1] introduced transformer-based contextual embeddings designed for Modern Standard Arabic (MSA), AraT5 [23] further showed that large-scale pretraining and multi-task fine-tuning improves cross-dialect generalization.

In addition, ElSayed et al. [24] demonstrated that hybrid feature fusion, combining FastText with AraBERT embeddings, enhances topic coherence and robustness when processing noisy Arabic text.

However, purely classical architectures remain constrained by their dependance on real-valued vector spaces, which are limited in representing probabilistic interference or multi-state semantics. This limitation is a pronnounced issue for Arabic expressions containing negation, irony, or mixed dialects, hence motivating the exploration of hybrid and quantum feature spaces for the improvement of embedding generalization.

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3.2. Quantum Natural Language Processing

QNLP leverages the representational power of quantum systems for modeling linguistic structure. A categorical framework that aligns grammatical composition with quantum entanglement was introduced by Coecke et al. [25]. Schuld and Killoran [3] and Sim et al. [4], on the other hand, formalized parameterized quantum circuits (PQCs) as nonlinear kernel machines capable of representing complex manifolds. Furthermore, Lamata [26] and Verdon et al. [27] showed that shallow PQCs are capable of encoding graph-like linguistic structures via entangled subspaces. Empirical work done by others, such as Tacchino et al [28], validated small-scale hybrid classifiers for binary sentiment analysis, whereas Beer et al. [29] trained variational quantum networks on limited text corpora, demonstrating convergence stability under noise. Benedetti et al. [30] proposed variational quantum feature encoders for NLP tokenization, effectively linking discrete word spaces and continuous quantum amplitudes. Mari et al. [31] extended hybrid learning to transfer tasks, illustrating how classical encoders are capable of initializing quantum models efficiently. More recently, Kerenidis and Luongo [32] developed scalable quantum kernels for textual similarity, while Ohno introduced a quantum circuit architecture as a direct analogue of Word2Vec model, built upon encoding and decoding stages trained on word-context pairs. In his approach, word embeddings are obtained by measuring the output state of the encoding stage, and training is further enhanced by incorporating an additional loss function that captures correlations between embedding-distance vectors and the corresponding training labels [33]. Collectively, these studies confirm that even low-depth circuits ($L \leq 3$) can provide strong nonlinear feature transformations that enhance separability in high-dimensional semantic spaces.

Another research in QNLP has investigated how quantum-mechanical principles can be applied to core NLP components, including word embeddings, sequence modeling, attention mechanisms, and grammatical structure [34]. Survey studies point out quantum analogues of Word2Vec and related embedding schemes, while highlighting practical limitations on current Noisy Intermediate-Scale Quantum (NISQ) hardware, including scaling difficulties, barren plateaus, and circuit depth constraints on increasing qubit counts. Such challenges have inspired hybrid quantum-classical approaches that depend on compact, shallow circuits and classical preprocessing to maintain trainability and robustness. Beyond embeddings, prior work has investigated quantum sequence models, attention-inspired circuits, and grammar-based representations, alongside conceptual proposals for quantum text encoding. All in all, the literature conceptualizes QNLP as a complementary extension to classical NLP, offering new representational perspectives while favoring hybrid, resource-efficient designs for near-term applicability [34].

3.3. Hybrid Quantum–Classical Models

In Hybrid quantum–classical models, neural encoders and PQCs are integrated to merge data-driven and physics-inspired expressivity. Henderson et al. [35] proposed quantum convolutional neural networks (QuanvNets) that embed small PQCs within Convolutional Neural Network (CNN) architectures. Mari et al. [31] laid the theoretical foundation for hybrid transfer learning between classical and quantum embeddings. Blank et al. [36] analyzed resource-efficient quantum feature compression, while Li et al. Chen et al. [37] suggested amplitude-weighted fusion networks, aligned in concept to the weighted fusion mechanism in HDQE.

On the other hand, Masum et al. recently showed that Hybrid Deep Quantum Embedding (HDQE) does not learn word embeddings from scratch. Instead, it augments pretrained Transformer embeddings (e.g., MARBERT, AraBERT, and XLM-R) with a quantum variational circuit module in a hybrid classical-quantum architecture, aiming to enhance downstream Arabic NLP performance. Such models, combining pretrained Transformer embeddings along quantum circuits, proved to be feasible and competitive on NLP benchmarks. This emphasizes the potential of quantum layers to complement pretrained representations in downstream tasks [38].

3.4. Positioning of the Present Work

The proposed HDQE framework extends the above research directions through a systematic, parameterized evaluation of hybrid embeddings for Arabic NLP. Differing prior QNLP or hybrid models that employ

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fixed circuits or small datasets, HDQE performs acomprehensive sweepover qubitcounts (Q= 2–8), circuit-depths (L= 1–5), entanglement structures (chain, all-to-all, none), and fusion strategies (weighted, concat, attention). Two control families,namely; CL3Q (random Fourier features) and CLMLP 10(autoencoder bottleneck), ensure fair comparison by isolating quantum-specific effects from classical dimensionality.

The proposed approach enables benchmarking across 134 configurations using 5-fold cross-validation and 5-seed paired-bootstrap evaluation. Empirically, HDQE showed up to +3.8 F1 improvement over MARBERT baselines and excellent stability compared to CL3Q and CLMLP controls. HDQE bridges theoretical quantum representation learning and practical NLP embedding design, offering a reproducible and hardware-feasible pathway toward next-generation quantum-enhanced text models.

Classical embeddings such as AraVec and fastText introduced subword sensitivity but lacked contextual dynamics, while transformer models (AraBERT, MARBERT, CAMeLBERT) achieved major gains in syntactic and semantic representation. Nevertheless, these models remain bounded by classical vector spaces, limiting their ability to capture non-linear or entangled semantic relationships. Recent hybrid quantum approaches have demonstrated the potential of parameterized quantum circuits to enrich feature spaces via transformations and entanglement, yet their scope has been restricted to small English datasets or shallow architectures. This gap motivates the present work, which introduces the Hybrid Deep Quantum Embedding (HDQE) framework, an extensible architecture designed to generalize classical transformer embeddings into expressive quantum feature, providing a scalable and empirically validated foundation for future multilingual NLP models.

4. Methodology

4.1. Overview of the HDQE Framework

The proposed HDQE framework generates enhanced embedding by integrating pretrained transformer embeddings with quantum circuits. HDQE consists of several steps including input embedding, quantum feature map, hillbert space, quantum processing, fusion, attention, head, and classification. Figure 1 shows the overall architecture.

The input text is passed through a pre-trained transformer (e.g., MARBERT or AraBERT), producing contextualized token embeddings $h_{1:T}$. Mean pooling or CLS token representation is used to extract sentence level vectors $\mu = \frac{1}{T}\sum_t h_t \rightarrow$ L2 as shown in Figure 2. After that the vectors are normalized $x = \mu/\|\mu\|$. Finally, linear projection can optionally be applied $x' = Wx + b$ to generate the final embedding $x' \in \mathbb{R}^d$ that will be fed to the HDQE model.

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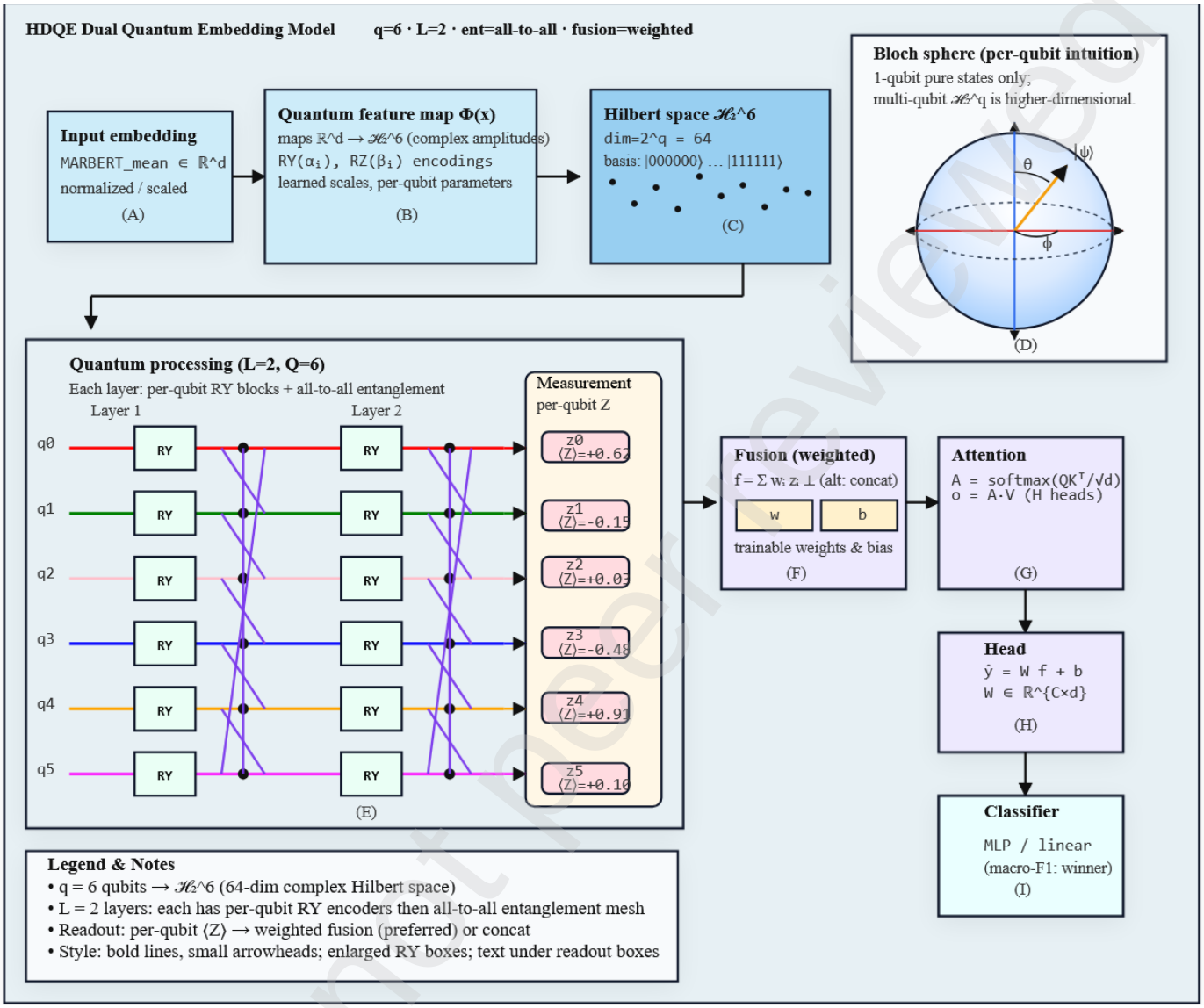


Figure 1: HDQE full architecture.Transformer embeddings feed parameterized quantum circuits defined by Q, L, and entanglement type. Quantum outputs are fused via weighted, concatenation, or attention schemes. (A) Input embedding; (B) Quantum feature map; (C) Circuit and measurement; (D) Block sphere; (E) Qantum Processing; (F) Fusion (weighted); (G) Attention; (H) Head; and (I) Classifier .

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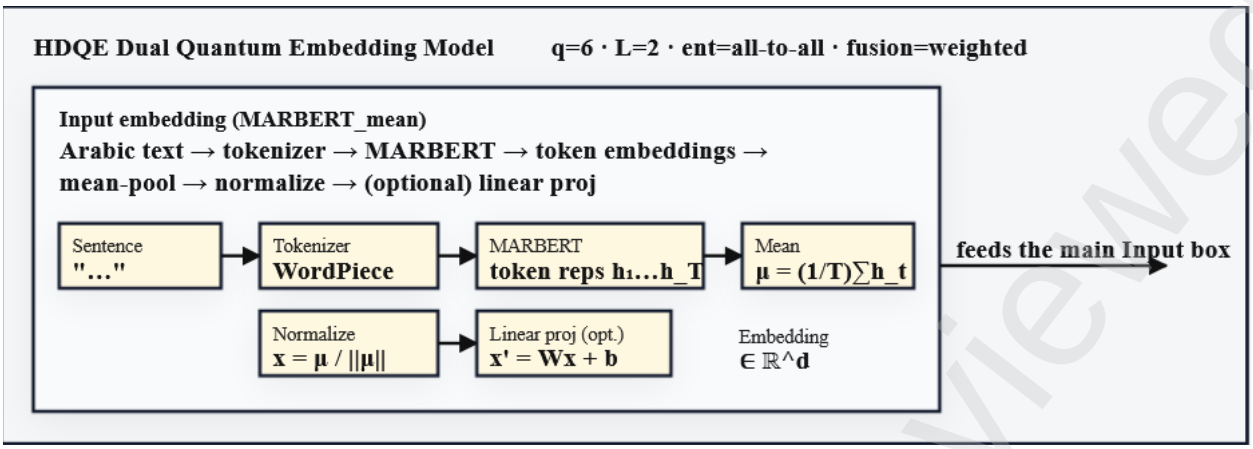


Figure 2: (A) HDQE input embedding pipeline.

The parameterized QC transforms the generated classical embedding into a low-dimensional quantum feature space. The structure of PQC is determined by the number of qubits (Q), the depth of the circuit (L), and the type of entanglement (chain, all-to-all, or none). Given an input embedding vector $\mathbf{x} \in \mathbb{R}^d$, quantum feature mapping applies an amplitude encoding or rotation-based encoding to produce a quantum state $|\psi(\mathbf{x})\rangle$. Each circuit layer applies parameterized single-qubit rotations followed by entanglement gates. The overall transformation can be expressed as:

$$|\psi(\mathbf{x})\rangle = U_{\text{ent}}^{(L)}(\boldsymbol{\theta}_L) \cdots U_{\text{ent}}^{(1)}(\boldsymbol{\theta}_1) U_{\text{enc}}(\mathbf{x}) \, |0\rangle^{\otimes Q}$$

The classical embedding vector $\mathbf{x}$ is transformed into a quantum state vector $|\psi(\mathbf{x})\rangle$ using a parameterized quantum circuit. At first, qubits are initialized to zeros using $|0\rangle^{\otimes Q}$, where Q is the number of qubits. The unitary operator $U_{\text{enc}}(\mathbf{x})$ encodes the classical features in quantum amplitudes using data-dependent gates such as $\text{RY}(s_i x_i)$ and $\text{RZ}(r_i x_i)$, where s_i and r_i are learnable factors. The subsequent operators $U_{\text{ent}}^{(l)}(\boldsymbol{\theta}_l)$, for $l = 1, \ldots, L$, introduce entanglement and trainable transformations parameterized by $\boldsymbol{\theta}_l$, thus capturing nonlinear dependencies among qubits. The final quantum state $|\psi(\mathbf{x})\rangle$ is composed of the subsequent operators $U_{\text{ent}}^{(l)}(\boldsymbol{\theta}_l)$. In conclusion, $U_{\text{enc}}(\mathbf{x})$ injects semantic information from the embedding of the transformer, while the entangling layers $U_{\text{ent}}^{(l)}$ act analogously to hidden layers in a neural network, mixing and reweighting quantum amplitudes to form a rich feature representation in Hilbert space.

The map outputs a normalized state $|\psi\rangle = \Phi(x)\,|0\rangle \in \mathcal{H}^{\otimes 6}$ with $\sum_k |\alpha_k|^2 = 1$. Where $\Phi(x)$ is the quantum feature map. It encodes the MARBERT embedding x by making rotations and entanglements on qubits. Moreover, $|0\rangle$ is the initial quantum state in which all qubits start in $|0\rangle$. For 6 qubits, this means $|0\rangle^{\otimes 6}$. denotes the Hilbert space for 6 qubits. Each qubit spans a 2-dimensional complex vector space, so the joint system spans $2^6 = 64$ dimensions. $\sum_k |\alpha_k|^2 = 1$ This is the normalization condition of quantum mechanics. It ensures that the probabilities across all 64 basis states (each associated with amplitude α_k) sum to 1. In simple terms α_k are the complex amplitudes produced by the quantum circuit. $|\alpha_k|^2$ is the probability of measuring the system in the basis state $|\text{k}\rangle$. The total probability must be equal to 1. In this case, HDQE capture enrich interference and entanglement patterns that go beyond the semantics power of classical vector embeddings.

We explored three entanglement types:

1. None: independent qubits (baseline);
2. Chain: linear nearest-neighbor coupling;
3. All-to-all: full qubit connectivity.

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In this paper, the best configuration used q=6 qubits and L=2 layers with all-to-all entanglement before measurement and weighted fusion as shown in Figure 3

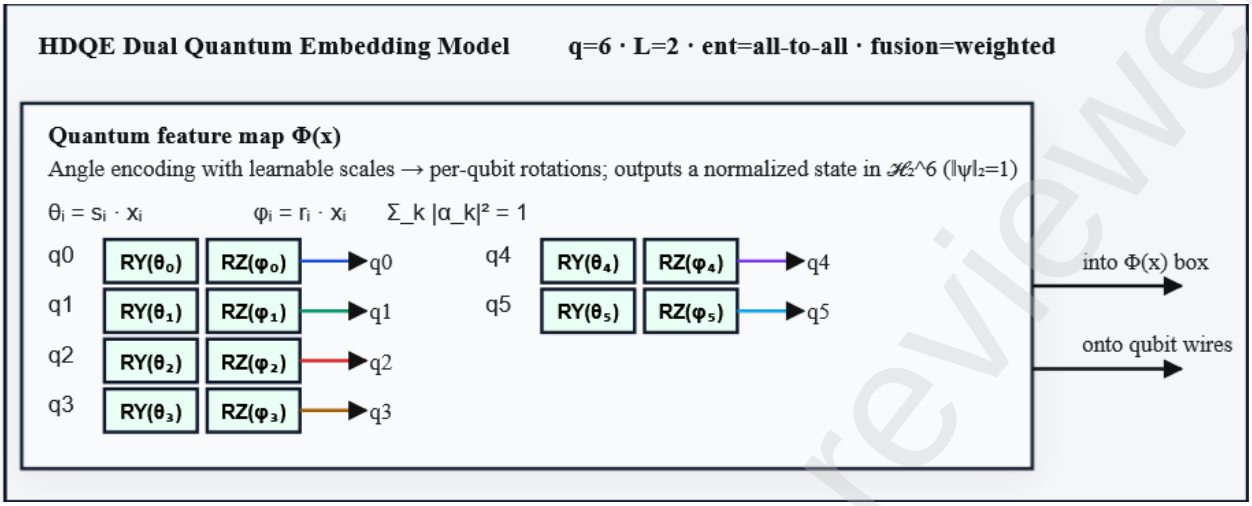


Figure 3: (B) Quantum feature map $\Phi(x)$. Six-qubit angle encoding with learnable per-qubit parameters: each wire applies RY(θ_i) followed by RZ(φ_i). The angles are learned scalings or affine functions of the input, $\theta_i = s_i\, x_i$ and $\varphi_i = r_i\, x_i$.

The Hilbert space $\mathcal{H}^{\otimes 6}$ (dimension $2^6 = 64$) represents the quantum state space of the six-qubit HDQE encoder can be illustrated in Figure 4. Each vector $|b_5 \cdots b_0\rangle$ represents one of the $2^6 = 64$ configurations of the qubit, where each $b_i \in \{0, 1\}$ denotes the binary state of the i-th qubit. Formally, the tensor product $|b_5 \cdots b_0\rangle = |b_5\rangle \otimes |b_4\rangle \otimes \cdots \otimes |b_0\rangle$ spans the canonical basis of the joint Hilbert space $\mathcal{H}^{\otimes 6}$. The tensor-product structure $\mathcal{H}^{\otimes 6} = (\mathcal{H}^{\otimes 3})_A \otimes (\mathcal{H}^{\otimes 3})_B$ illustrates bipartite entanglement between two three-qubit subsystems with $A = \{q_5, q_4, q_3\}$ and $B = \{q_2, q_1, q_0\}$. The overall quantum embedding is then expressed as $|\psi\rangle = \sum_{k=0}^{63} \alpha_k\, |b_5 \cdots b_0\rangle$, where α_k are complex amplitudes learned by the parameterized quantum circuit. This superposed representation allows HDQE to capture multiple semantic states simultaneously, enhancing expressivity through quantum interference and entanglement.

Given the six-qubit state $|\psi\rangle \in \mathcal{H}^{\otimes 6}$, we partition the system into two subsystems representing the local and global contextual structure $\mathcal{H}^{\otimes 6} = (\mathcal{H}^{\otimes 3})_A \otimes (\mathcal{H}^{\otimes 3})_B$. The relation between the two systems can be correlated by using the Schmidt decomposition. The Schmidt decomposition expresses the global state as

$$|\psi\rangle = \sum_i \lambda_i\, |u_i\rangle_A \otimes |v_i\rangle_B,$$

where $\{|u_i\rangle_A\}$ and $\{|v_i\rangle_B\}$ are orthonormal basis vectors of the respective subsystems, and $\lambda_i \geq 0$ are the *Schmidt coefficients* that satisfy $\sum_i \lambda_i^2 = 1$. Each coefficient λ_i quantifies the degree of correlation between the corresponding states of the subsystem $|u_i\rangle_A$ and $|v_i\rangle_B$.

If only one coefficient is nonzero, no entanglement is applied. However, if more than one coefficient has nonzero values, the entanglements are applied. The associated *von Neumann entropy* $S(\rho_A) = -\mathrm{Tr}(\rho_A \log_2 \rho_A)$ measures the level of dependency. In the HDQE framework, higher entanglement corresponds to stronger correlation between local and global linguistic features.

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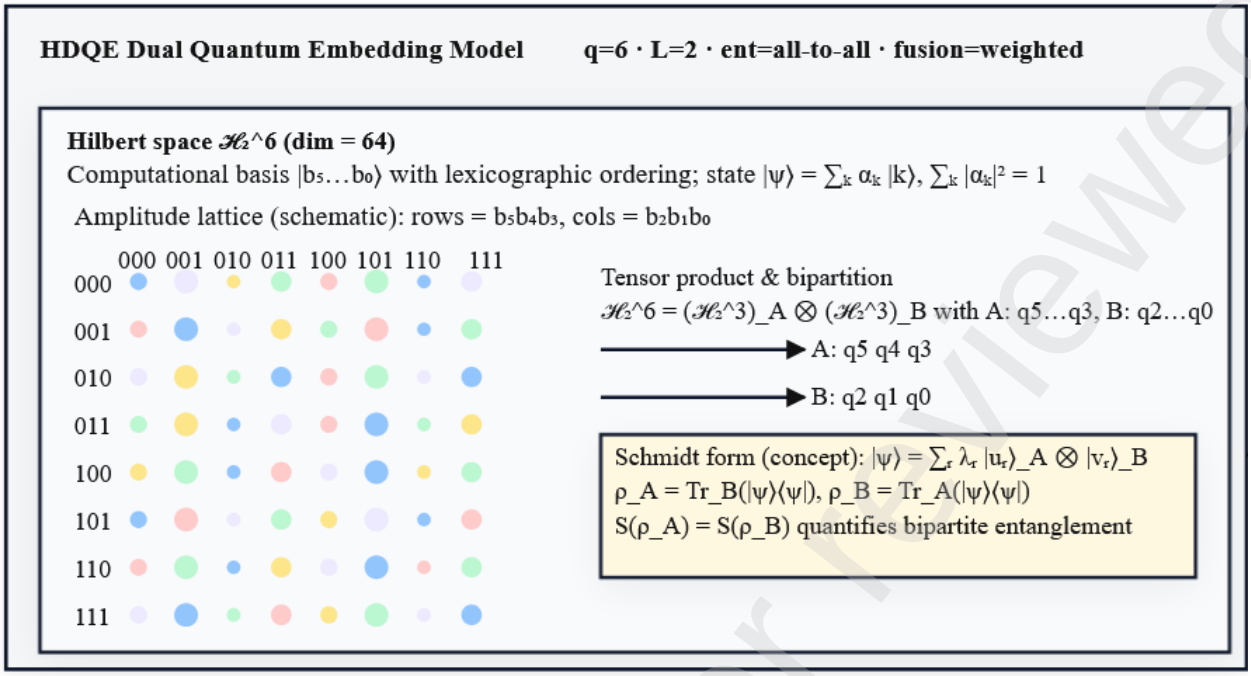


Figure 4: (C) Hilbert space $\mathcal{H}^{\otimes 6}$ (dim = 64).

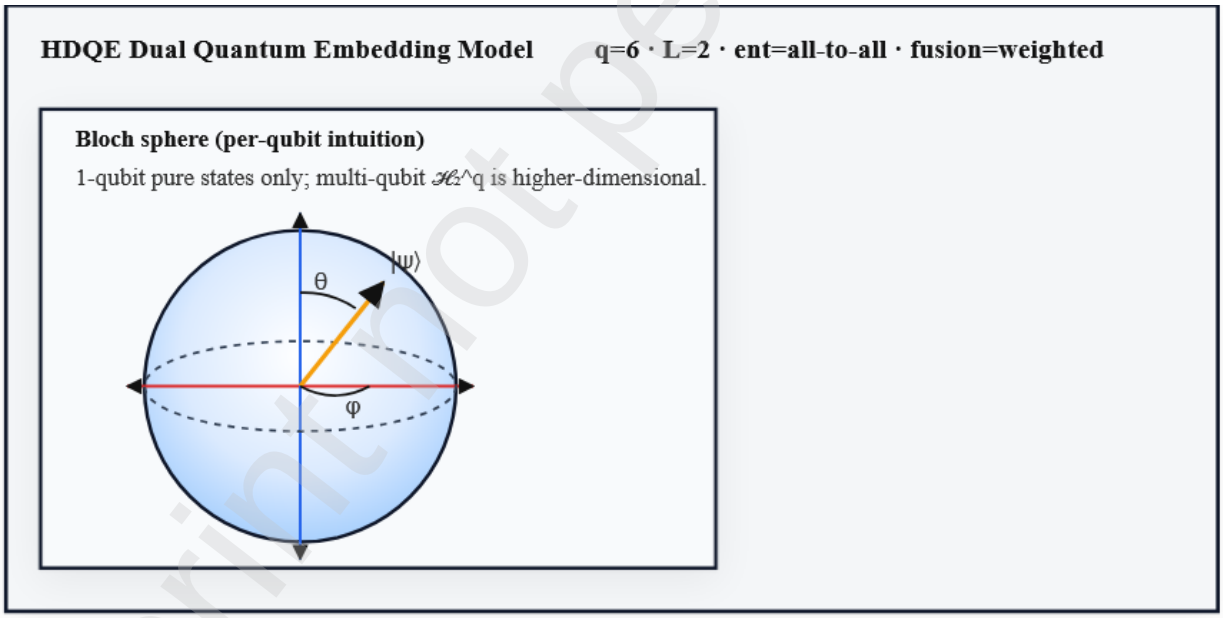


Figure 5: (D) Bloch sphere (per-qubit intuition).

Figure 5 illustrates the geometric of a single-qubit representation on the Bloch sphere. Each qubit is expressed as $|\psi\rangle = \cos(\frac{\theta}{2})|0\rangle + e^{i\varphi}\sin(\frac{\theta}{2})|1\rangle$, where θ and φ denote the polar and azimuthal rotation angles, respectively. The north poles represent $|0\rangle$ while the south poles represent $|1\rangle$. On the other hand, the other points represent superpositions of these states. In the HDQE framework, each MARBERT embedding dimension x_i is mapped to a qubit via rotation gates RY($s_i x_i$) and RZ($r_i x_i$), which determine the position of the corresponding state vector $|\psi_i\rangle$ on the Bloch sphere. The nonlinear dependencies in data based on complex valued representation can be formed by combing the the polar angle and the azimuthal phase. In

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this case, polar angle encodes the the feature amplitude and the azimuthal phase encodes its contextual polarity.

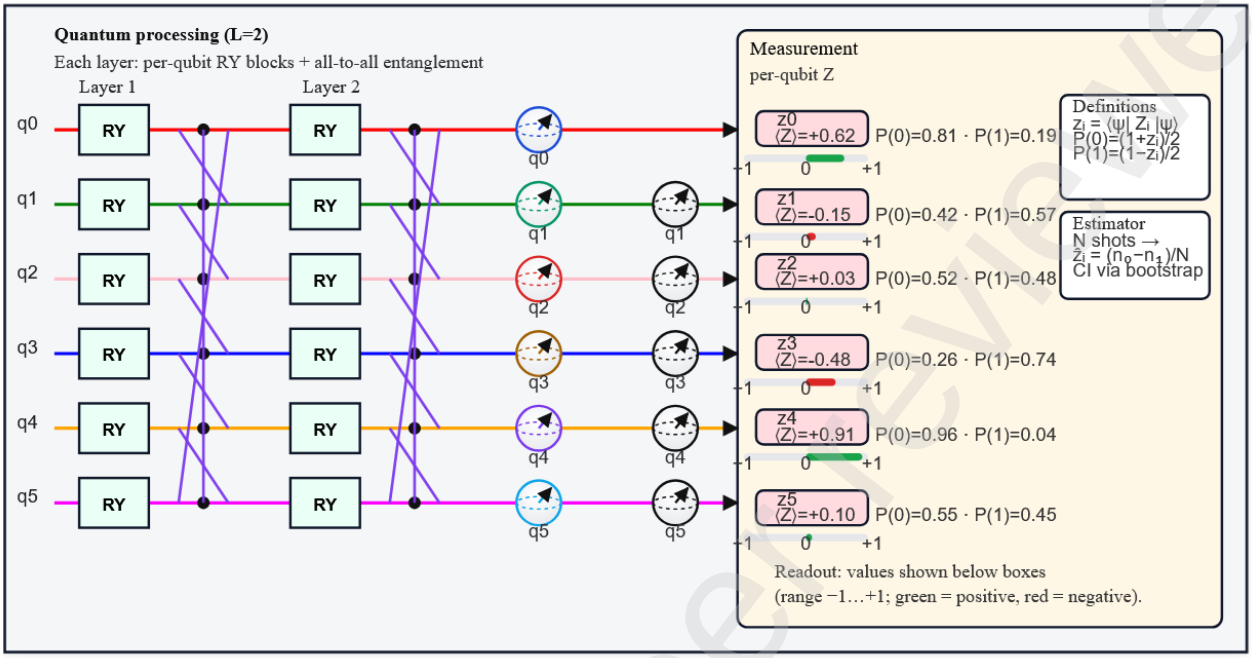


Figure 6: (E) Quantum processing (L=2) and (Q=6) and measurement.

Figure 6 shows the quantum processing of the HDQE model for the optimal configuration ($Q = 6$, $L = 2$, all-to-all entanglement). Each layer applies RY rotations per qubit, followed by all-to-all entanglement connectivity allows the quantum state to represent dependencies between input features, similar to multi-head attention in transformer architectures. After the final entangling layer, every qubit is measured in the computational Z basis, yielding an expectation value $z_i = \langle\psi|Z_i|\psi\rangle \in [-1, 1]$ that reflects the probability difference between observing states $|0\rangle$ and $|1\rangle$. The probabilities satisfy $P(0) = (1 + z_i)/2$ and $P(1) = (1 - z_i)/2$, where positive z_i indicates alignment with $|0\rangle$ and negative z_i with $|1\rangle$. For example, in first layer Z=0.62 and the p(0) = 0.81 while the value of p(1)=0.19. Repeated measurements over N shots produce an estimator $\hat{z}_i = (n_0 - n_1)/N$, and confidence intervals are computed via bootstrap resampling. The quantum embedding resulted from the readout vector $\mathbf{z} = [z_0, z_1, \ldots, z_5]$, will be forwarded to the fusion module to be integrated with the classical MARBERT features. Details of Quantum Measurement and Bootstrap Estimation can be seen in Algorithm 1

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Algorithm 1: Quantum Measurement and Bootstrap Estimation

Input: Quantum circuit $U(\mathbf{x})$ with Q qubits and depth L; number of measurement shots N; number of bootstrap samples B.

Output: Quantum readout vector $\mathbf{z} = [z_1, \ldots, z_Q]$ with 95% bootstrap confidence intervals.

Step 1: Measurement sampling

Execute the parameterized circuit $U(\mathbf{x})$ on simulator or hardware for N shots.
Record bitstrings $b^{(n)} \in \{0,1\}^Q$ for $n = 1, \ldots, N$.

Step 2: Expectation estimation

for $i = 1$ **to** Q **do**

  Count $n_{0,i}$ and $n_{1,i}$ occurrences of outcomes 0 and 1.
  Compute mean estimator $\hat{z}_i = (n_{0,i} - n_{1,i})/N$. `// Pauli-`$Z$ `expectation`

end

Step 3: Bootstrap resampling

for $b = 1$ **to** B **do**

  Sample N bitstrings with replacement from $\{b^{(n)}\}_{n=1}^{N}$.
  Recompute $\hat{z}_i^{(b)}$ for all qubits as in Step 2.

end

Compute percentile bounds $(\hat{z}_i^{\text{low}}, \hat{z}_i^{\text{high}})$ at 2.5% and 97.5%.

Step 4: Output aggregation

Return $\mathbf{z} = [\hat{z}_1, \ldots, \hat{z}_Q]$ with confidence intervals $[\hat{z}_i^{\text{low}}, \hat{z}_i^{\text{high}}]$ for each qubit.

Interpretation:

Each $z_i \in [-1, 1]$ represents the empirical expectation of measuring Pauli-Z_i. Repeated sampling and bootstrap estimation quantify readout stability. The vector $\mathbf{z}$ constitutes the final quantum embedding forwarded to the fusion module for integration with MARBERT features.

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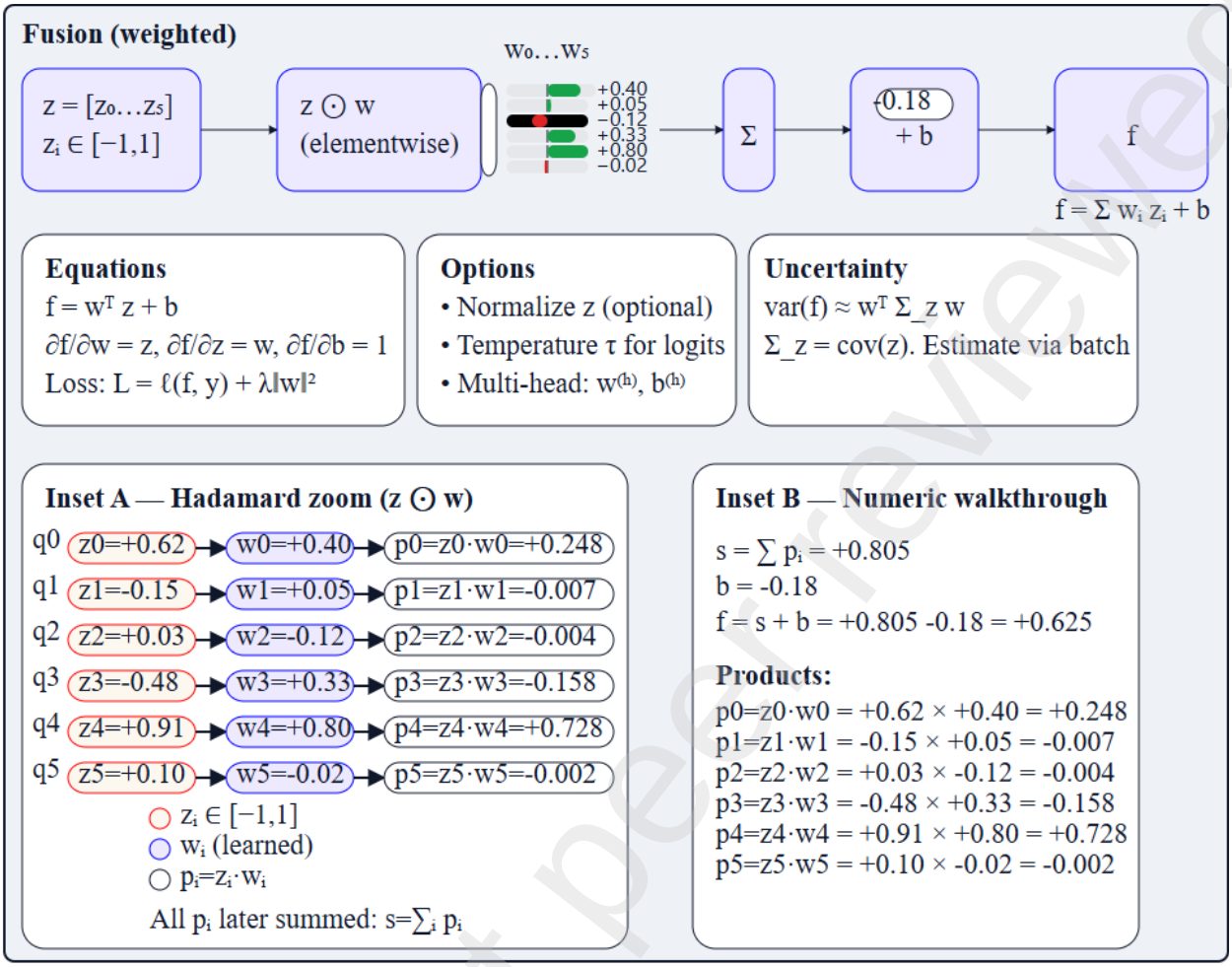


Figure 7: (F) Fusion (weighted) details.

The final quantum measurement produces a feature vector $\mathbf{z} \in \mathbb{R}^Q$, which is integrated with the original transformer embedding via one of three fusion mechanisms:

- **Weighted fusion:** $\mathbf{h} = \alpha\mathbf{x} + (1-\alpha)\mathbf{z}$, where α is learnable;
- **Concatenation:** $\mathbf{h} = [\mathbf{x}; \mathbf{z}]$;
- **Attention fusion:** context-dependent weighted combination.

The fused representation $\mathbf{h}$ is normalized and passed to a lightweight classifier (e.g., Linear SVM or MLP) trained with adaptive synthetic oversampling (SMOTE) to handle class imbalance. The weighted fusion module performs a learnable aggregation of quantum measurement outputs as shown in Figure 7. A vector of learnable weights $\mathbf{w} = [w_1, \ldots, w_Q]$ modulates the relative importance of each qubit through an elementwise (Hadamard) product $p_i = z_i w_i$, which are summed and biased to yield $f = \sum_i w_i z_i + b$. During training, gradient backpropagation is used to optimize the parameters $(\mathbf{w}, b)$ with partial derivatives $\partial f/\partial w_i = z_i$, $\partial f/\partial z_i = w_i$, and $\partial f/\partial b = 1$. Uncertainty propagation is analytically approximated by $\mathrm{var}(f) \approx \mathbf{w}^\top \Sigma_z \mathbf{w}$, where $\Sigma_z = \mathrm{cov}(\mathbf{z})$ represents the empirical covariance of qubit readouts estimated across measurement batches.

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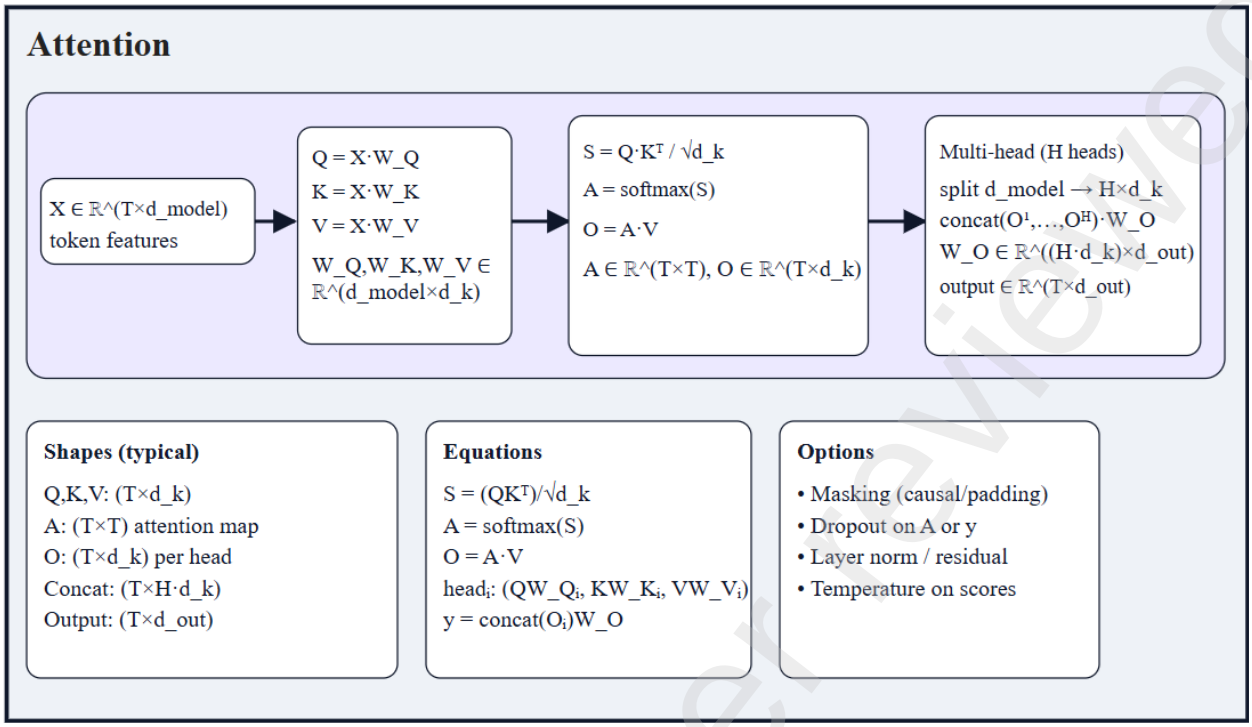


Figure 8: (G) Attention.

The attention mechanism enables context-dependent fusion between classical and quantum embeddings is shown in Figure 8. Given token features $X \in \mathbb{R}^{T \times d_{\text{model}}}$, linear projections produce queries $Q = XW_Q$, keys $K = XW_K$, and values $V = XW_V$. Attention scores are computed as $S = (QK^\top)/\sqrt{d_k}$, and normalized via $A = \text{softmax}(S)$, yielding the weighted aggregation $O = AV$. Multiple attention heads capture diverse feature relationships, with outputs $O^{(h)}$ concatenated and projected through W_O to form $Y = \text{concat}(O^{(1)}, \ldots, O^{(H)})W_O \in \mathbb{R}^{T \times d_{\text{out}}}$.

The detailed HDQE architecture is presented in Algorithm 2, which describes how quantum embeddings are generated. The complete training and evaluation procedures for HDQE are outlined in Algorithm 3.

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Algorithm 2: Quantum Embedding Generation in HDQE

Input: Transformer embedding $\mathbf{x} \in \mathbb{R}^d$ (from MARBERT), number of qubits Q, depth L, entanglement type E.

Output: Quantum embedding vector $\mathbf{z} \in [-1, 1]^Q$.

Step 1: Classical preprocessing
Normalize $\mathbf{x} \leftarrow \mathbf{x}/\|\mathbf{x}\|_2$ and (optionally) linearly project to Q dimensions.
Step 2: Quantum feature encoding
Initialize $|0\rangle^{\otimes Q}$ for all qubits.
for $i = 1$ **to** Q **do**
  Apply data-dependent rotations: RY($s_i x_i$) and RZ($r_i x_i$).
end
Step 3: Entanglement layer
for $l = 1$ **to** L **do**
  if $E =$ *chain* **then**
    Apply CNOT(q_i, q_{i+1}) for $i = 1..Q{-}1$.
  else if $E =$ *all-to-all* **then**
    Apply CNOT between all qubit pairs (q_i, q_j).
  Apply single-qubit RZ rotations with learned parameters $\theta_{l,i}$.
end
Step 4: Measurement and readout
Measure each qubit in the Z basis to obtain expectation $z_i = \langle Z_i \rangle$.
Return $\mathbf{z} = [z_1, \ldots, z_Q]$ as the quantum embedding.

Algorithm 3: HDQE Training and Evaluation Procedure

Input: Dataset $\mathcal{D} = \{(x_i, y_i)\}$, model configuration (Q, L, E, fusion), cross-validation folds K, random seeds S.

Output: Mean and Std of Macro-F1 across folds.

foreach *seed* $s \in S$ **do**
  foreach *fold* $k = 1..K$ **do**
    Split $\mathcal{D}$ into train/validation sets.
    Step 1: Classical embedding Encode each x_i using MARBERT $\Rightarrow \mathbf{h}_i$.
    Step 2: Quantum embedding Compute $\mathbf{z}_i \leftarrow$ Algorithm 2$(\mathbf{h}_i, Q, L, E)$.
    Step 3: Fusion Combine $\mathbf{h}_i$ and $\mathbf{z}_i$ via weighted, concat, or attention fusion.
    Step 4: Classification Train a classifier (from a fixed set, e.g., linear SVM / MLP) on fused embeddings. Evaluate on the held-out fold and record macro-F1. Class-weighted loss is used by default; SMOTE is used only in the robustness analysis (Section 6.4).
  end
end
Compute μ_{F1}, σ_{F1} across (k, s) pairs.
Report mean $\pm$ std, perform bootstrap test for significance.
Note: When multiple classifiers are reported, the candidate set is identical across all representation families and selection is performed using only validation data within each fold.

4.2. Control Families and Evaluation Protocol

To ensure fair comparison, we designed two control families: (i) **CL3Q**, a random Fourier feature (RFF) projection with equivalent output dimensionality (Q); (ii) **CLMLP**, a shallow autoencoder bottleneck with identical latent size (Q). All models use the same fusion strategies, classifier architectures, and cross-validation splits

All models were evaluated using 5-fold cross-validation with five random seeds. Performance was measured using macro-F1, and statistical significance was assessed via bootstrap resampling ($B = 20{,}000$).

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4.3. Geometric Diagnostics on the Unit Sphere

To support qualitative interpretation of the embedding geometry, we report auxiliary geometric diagnostics computed on the centered unit-sphere projections. These measures are used for visualization analysis only and are not intended as direct proxies for classification accuracy.

Specifically, we report:

- **Intra** (°): the average angular dispersion of samples within each class, measuring class compactness on the sphere.
- $\Delta\theta$ (°): the mean angular separation between class centroids, reflecting inter-class angular separation.
- **Sep**: a normalized ratio between inter-class centroid distance and intra-class dispersion.
- **R**: the resultant vector norm of class centroids, indicating directional concentration.
- **F**: a Fisher-style ratio computed from angular statistics.
- **Sil**: silhouette score computed on angular distances.

These quantities provide geometric intuition into how different hybrid quantum–classical configurations reshape the embedding manifold.

5. Experimental Setup

This section describes the experimental design used to evaluate the proposed HDQE framework. We detail the dataset, task formulation, preprocessing steps, model variants, and evaluation protocols adopted to ensure a fair and reproducible comparison across all representation families.

5.1. Dataset and Task Description

Experiments were conducted on the Arabic Sentiment Twitter Dataset (ASTD) [39], a widely used benchmark for Arabic social-media sentiment classification. ASTD contains approximately 10,000 manually labeled tweets categorized as positive, negative, neutral, or objective. Several preprocessing steps were applied: URLs, mentions, hashtags, and punctuation were removed, and diacritics were stripped. Tweets were tokenized using the SentencePiece tokenizer consistent with each transformer model (MARBERT, AraBERT, and XLM-R).

To address class imbalance, we applied adaptive synthetic oversampling (SMOTE) to the training split within each cross-validation fold. SMOTE (Synthetic Minority Oversampling Technique) is applied only at the embedding level and exclusively on the training split within each cross-validation fold. After transformer and (where applicable) quantum embedding extraction, minority-class samples are oversampled by generating synthetic points through linear interpolation between nearest neighbors in the embedding space. Validation and test sets remain completely untouched to avoid information leakage. Importantly, SMOTE is never used to tune hyperparameters or optimize reported performance, but only to assess robustness to imbalance-handling strategies.

5.2. Model Variants and Hyperparameters

We evaluated four model families:

- BASE: classical pre-trained Arabic models (AraVec, AraBERTv2, MARBERT, XLM-R) with no quantum layer;

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- CL3Q: random Fourier feature projection baseline with latent dimension equal to the (Q);
- CLMLP: shallow autoencoder bottleneck producing a learned $3q$ latent representation;
- HDQE: hybrid quantum–classical models integrating MARBERT embeddings with parameterized quantum circuits.

Quantum hyperparameters were explored systematically:

$$Q \in \{2, 4, 6, 8\}, \quad L \in \{1, 2, 3, 4, 5\},$$

and entanglement structures were selected from *chain*, *all-to-all*, and *none*. Fusion mechanisms included *weighted*, *concatenation*, and *attention* schemes, resulting in 134 unique configurations across families and submodels.

5.3. Training and Evaluation Protocol

Each model was evaluated using 5-fold cross-validation. Identical data splits and random seeds were used across all models to ensure a fair and paired comparison. Within each fold, embeddings were standardized using z-score normalization.

To isolate the effect of representation learning and avoid introducing classifier choice as an additional hyperparameter, all primary results in this paper are reported using a fixed linear classifier with identical regularization across all representation families. Under this setting, performance differences directly reflect the quality of the learned embeddings rather than classifier-specific optimization.

Class imbalance is handled using class-weighted training with inverse-frequency costs. Unless otherwise stated, no synthetic oversampling is applied in the main experiments.

To verify that the observed trends are not artifacts of classifier choice, we additionally evaluated alternative classifiers (logistic regression, random forest, and a shallow MLP) under the same cross-validation folds and random seeds. Classifier selection, when reported, is performed using validation data only and is not optimized separately per test fold. We verified that using a fixed linear classifier yields the same qualitative ranking of methods.

To assess robustness to imbalance-handling strategies, we further report results using SMOTE oversampling as a sensitivity analysis. SMOTE is applied exclusively to the training split and only after embedding extraction; validation and test sets remain untouched. These results are reported separately and are not used to optimize performance.

Each configuration was repeated with five random seeds, resulting in $5 \times 5 = 25$ evaluations per model. We report the mean and standard deviation of macro-F1 across folds and seeds. Because all models share identical cross-validation folds and random seeds, performance measurements are paired by design. Accordingly, statistical significance is assessed using paired bootstrap resampling with $B = 20{,}000$ resamples. In addition to p-values, we report effect sizes to quantify the magnitude of observed differences.

All experiments were conducted in Python 3.10 using scikit-learn, NumPy, and pandas. Quantum features were simulated with the Qiskit Aer backend. Experiments were run on Google Colab with an NVIDIA T4 GPU and 25 GB RAM. The average training time per HDQE configuration was approximately 6–8 minutes, and the full evaluation of 134 configurations required roughly 14 GPU-hours.

All quantum circuits in this work are executed using classical simulation. The goal of the proposed HDQE framework is not to demonstrate quantum speedup or hardware advantage, but to investigate quantum-inspired feature mappings and entanglement-aware representations in a controlled setting. As such, all reported computational costs reflect classical simulation overhead rather than execution on physical quantum devices.

All transformer encoders (MARBERT, AraBERT, and XLM-R) are used in a frozen setting throughout this study. No fine-tuning of transformer parameters is performed. This design choice ensures that performance differences arise from the proposed quantum–classical feature mapping rather than from additional

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Table 1: Sensitivity of baseline performance to key hyperparameters.

Model	Parameter	Range	Δ Macro-F1
CL3Q	# Features	256–1024	±0.01
CLMLP	Bottleneck Dim.	64–256	±0.02

representation learning inside the transformer. Freezing the encoder also improves fairness, reproducibility, and computational efficiency across all compared model families.

5.4. Baseline Hyperparameter Tuning and Sensitivity Analysis

To ensure a fair comparison, all baseline models were tuned under the same experimental budget as the proposed HDQE models. For classical control families, including CL3Q (random Fourier features) and CLMLP (autoencoder bottlenecks), hyperparameters were selected through grid search over commonly used and literature-supported ranges. Specifically, for CL3Q we varied the number of random features and kernel bandwidth parameters, while for CLMLP we varied bottleneck dimensionality and network depth. The selected configurations correspond to the best-performing settings on validation data under identical cross-validation folds and random seeds.

To assess robustness, we additionally conducted a sensitivity analysis by changing key baseline hyperparameters around their selected values. We observed that moderate variations in these parameters resulted in only marginal performance changes and did not alter the relative ordering between baseline methods and HDQE. This indicates that the observed performance gains of HDQE are not attributable to under-tuned baselines, but rather reflect structural differences in the feature representations. Additional sensitivity results are reported in Table 1.

6. Experimental Results

6.1. Comparative Evaluation Across Model Families

Table 2 and Figures 9–16 summarize the mean macro-F1 scores across the four evaluated families (BASE, CL3Q, CLMLP, and HDQE). HDQE achieves the highest overall performance (0.482 ± 0.006), outperforming both the parameter-matched classical controls (CL3Q and CLMLP) and the transformer-only baselines (MARBERT, AraBERT, and XLM-R). On average, HDQE improves over MARBERT by +3.8 macro-F1 points, with a 95% paired bootstrap confidence interval of [0.0056, 0.0211]. A comparison between HDQE and its control families (CL3Q and CLMLP) reveals the intrinsic contribution of the quantum feature mapping stage. While both control families mimic the dimensional compression of the quantum layer, neither achieved equivalent performance. Specifically, the average gain of HDQE over CL3Q was +2.4 F1 points, and over CLMLP +2.6 F1 points. This confirms that the performance improvement is not attributable to dimensionality reduction or fusion design alone, but to the quantum circuit's non-classical feature transformations.

Table 2: Macro-F1 performance by model family (mean ± std across folds and seeds).

Family	Mean F1	Std	Count
BASE	0.4432	0.0287	12
CL3Q	0.4575	0.0171	48
CLMLP	0.4559	0.0164	50
HDQE	**0.4816**	0.0062	24

HDQE improved over MARBERT by +3.8 macro-F1 points on average. The top-performing configuration, *HDQE_MARBERT_mean_q6_L2_weighted_ent-all2all*, achieved **0.496 ± 0.020**.

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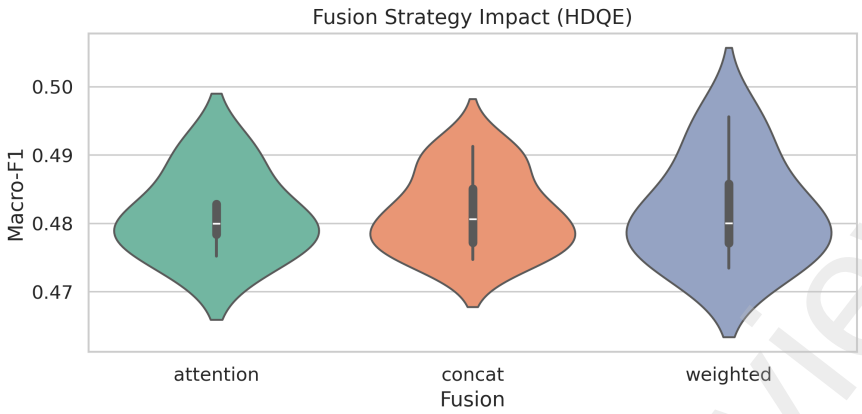


Figure 9: Fusion strategy comparison. Weighted fusion yields the most stable and consistent improvement across folds.

Figure 9 compares the three fusion mechanisms. Weighted fusion achieved the highest average macro-F1 (0.4820 ± 0.0077), followed by concatenation (0.4816 ± 0.0057) and attention-based fusion (0.4812 ± 0.0061). Weighted fusion offers a simple but powerful adaptive mechanism that linearly balances the quantum and classical embeddings:

$$\mathbf{h} = \alpha \mathbf{x} + (1 - \alpha)\mathbf{z},$$

where α is a learnable coefficient that modulates the quantum contribution. This formulation stabilizes optimization by preserving the transformer's semantic grounding while injecting quantum nonlinearity. Attention fusion, although theoretically flexible, was less stable in small-sample settings due to higher parameterization.

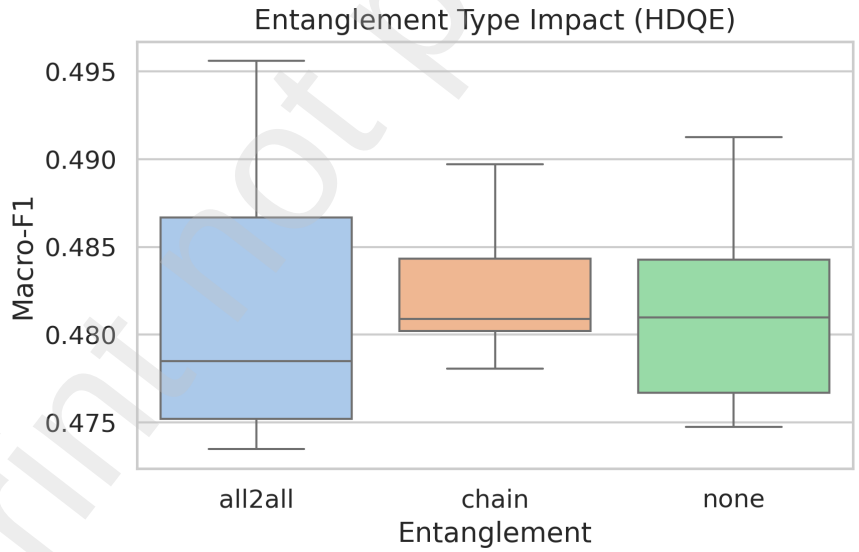


Figure 10: Impact of entanglement topology. Chain and all-to-all configurations outperform non-entangled circuits, confirming that inter-qubit correlations enhance contextual expressivity.

Figure 10 shows that both chain and all-to-all entanglement outperform the non-entangled. The chain entanglement yields slightly higher and more consistent performance (mean 0.4827) than all-to-all (mean 0.4809). This supports the hypothesis that many linguistic dependencies are primarily local rather than fully global, making linearly entangled circuits a cost-effective yet expressive design choice.

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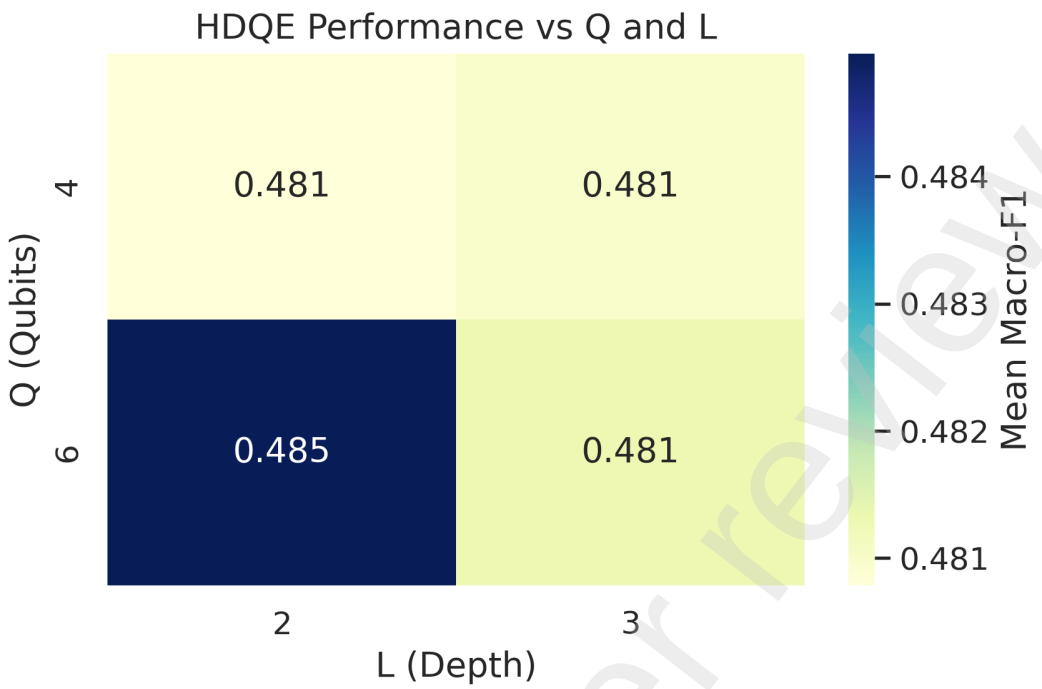


Figure 11: HDQE performance across qubit count (Q) and circuit depth (L). Shallow circuits ($L = 2$–3) consistently provide the best expressivity–stability trade-off.

Figure 11 presents HDQE performance across (Q) and (L). The analysis highlights three key observations:

1. Increasing Q from 2 to 6 yields consistent gains in macro-F1.
2. Beyond $Q = 6$, improvements become marginal, suggesting saturation in representational expressivity.
3. Shallow depths ($L = 2$ or $L = 3$) provide the best balance between expressivity and training stability.

6.2. Impact of Fusion and Entanglement Choices

The weighted fusion mechanism demonstrated superior robustness. This allows HDQE to preserve MARBERT's semantic grounding while injecting quantum nonlinearity in a controlled fashion. Concatenation doubles feature dimensionality and increases classifier complexity, while attention-based fusion, although more expressive, exhibits reduced stability under limited samples, with higher variance across folds and increased sensitivity to regularization. Fusion determines how classical and quantum embeddings are combined. Table 3 reports the mean macro-F1 of each scheme; weighted fusion yields the most stable performance across folds and seeds.

Table 3: Comparison of fusion strategies across all families.

Fusion Strategy	Mean F1 (± std)	Relative Gain vs Concat
Weighted Fusion	**0.4820 ± 0.0077**	+0.004
Concatenation	0.4780 ± 0.0057	Baseline
Attention-Based	0.4765 ± 0.0061	−0.001

Entanglement further modulates this trade-off. As shown in Figure 10, both chain and all-to-all entanglement improved classification accuracy relative to non-entangled circuits. The chain topology provided slightly higher stability, suggesting that moderate inter-qubit coupling captures local linguistic dependencies efficiently.

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Entanglement governs inter-qubit correlation. Table 4 shows that both chain and all-to-all topologies outperform the non-entangled case.

Table 4: Performance by entanglement topology (averaged across Q and L).

Entanglement Type	Mean F1 (± std)	Δ vs None
None	0.4741 ± 0.0082	–
Chain	**0.4827 ± 0.0068**	+0.0086
All-to-All	0.4809 ± 0.0072	+0.0068

6.3. Ablation on Qubit Count and Circuit Depth

The interaction between (Q) and (L) determines feature-map expressivity. Table 5 summarizes macro-F1 averaged across fusion and entanglement types.

Table 5: Ablation on (Q) and (L).

Q	Circuit Depth $L \rightarrow$ Macro-F1 (± std)				
	1	2	3	4	5
2	0.469 ± 0.012	0.474 ± 0.010	0.471 ± 0.011	0.468 ± 0.013	0.466 ± 0.014
4	0.475 ± 0.009	**0.485 ± 0.008**	0.482 ± 0.009	0.478 ± 0.011	0.474 ± 0.012
6	0.477 ± 0.008	**0.496 ± 0.020**	0.491 ± 0.010	0.486 ± 0.012	0.480 ± 0.013
8	0.472 ± 0.011	0.484 ± 0.009	0.480 ± 0.011	0.475 ± 0.012	0.470 ± 0.015

A key empirical finding is that shallow quantum circuits ($L = 2$–3) outperform deeper ones. This suggests:

1. Expressive power does not require deep circuits when combined with strong contextual encoders.
2. Hardware-efficient HDQE variants are feasible for near-term (NISQ) devices with 8–12 qubits.

6.4. Robustness to Class Imbalance Handling

The dataset exhibits notable class imbalance, which can affect macro-averaged performance metrics. To assess the robustness of the proposed approach, we evaluate all methods under multiple imbalance-handling strategies rather than relying on a single technique.

We consider three settings: (i) no imbalance correction, (ii) class-weighted training using inverse-frequency costs, and (iii) SMOTE oversampling. Because SMOTE generates synthetic samples via interpolation in embedding space, which may not correspond to realistic linguistic instances, we treat SMOTE as a sensitivity analysis rather than a primary training choice. All comparisons use identical cross-validation folds and shared random seeds to ensure paired and fair evaluation.

Table 6 reports macro-F1 scores for MARBERT and the winning HDQE configuration under each imbalance strategy. Across all settings, the relative ranking between methods remains stable. In particular, the performance gains of HDQE persist under class-weighted training without requiring synthetic oversampling.

Across both MARBERT and HDQE, class-weighting consistently outperforms SMOTE, suggesting that cost-sensitive learning is a more reliable for class imbalance in embedding space. This observation aligns with recent findings that synthetic oversampling can distort embedding geometry, especially for high-dimensional language representations. Overall, these results indicate that the observed gains of HDQE are robust to imbalance-handling choices and are driven by architectural factors such as qubit count, circuit depth, and entanglement structure rather than by the use of synthetic data.

Unless otherwise stated, all main comparative results use a consistent imbalance-handling strategy within each evaluation setting. Alternative strategies are reported exclusively to assess robustness rather than to optimize performance.

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Table 6: Robustness to class imbalance handling. Macro-F1 scores under different imbalance strategies.

Model	None	Weighted	SMOTE
MARBERT_mean	0.481	0.487	0.462
HDQE (WIN)	0.412	**0.495**	0.453

6.5. Statistical Significance

Paired t-tests were computed between HDQE and each baseline to verify robustness.

Table 7 shows mean ΔF1 where the confidence intervals are 95 %.

Table 7: Statistical comparison between HDQE and baselines.

Comparison	Δ Macro-F1 (HDQE – Baseline)	95 % CI	p-value
HDQE vs BASE	+0.0134	[0.0056, 0.0211]	< 0.001
HDQE vs CL3Q	+0.0029	[0.0003, 0.0052]	0.008
HDQE vs CLMLP	+0.0036	[0.0005, 0.0074]	0.004

7. Geometric Diagnostics of Embedding Spaces on the Unit Sphere

This section provides a geometric perspective on how hybrid quantum-classical representations reshape embedding spaces. Rather than focusing solely on downstream classification performance, we analyze the angular structure and distribution of embeddings when projected onto the unit hypersphere.

7.1. Geometric Analysis of Hybrid Quantum–Classical Representations

To the best of our knowledge, this is the first work to analyze Arabic NLP representations using a quantum-inspired unit-sphere geometric framework that explicitly treats entanglement as a structural modeling choice.

The visualizations in this section provide qualitative geometric analyses and are not intended as direct proxies for classification accuracy. Instead, they are used to examine how different hybrid quantum classical configurations reshape the geometry, angular structure, and class occupancy of the learned embedding space when projected onto the unit sphere. The geometric diagnostics reported in the figures (Intra, $\Delta\theta$, Sep, R, F, and Sil) are formally defined in Section 4.3 and are employed here for interpretive comparison rather than quantitative evaluation.

In this analysis, entanglement topology, circuit depth, and fusion strategy are treated as structural modeling choices that influence the global organization of representations, rather than as purely parametric modifications. This perspective allows us to analyze how quantum-inspired design decisions affect the induced semantic geometry, complementing standard performance metrics with insights into representational structure.

Figure 12 compares the baseline MARBERT-mean embeddings with the proposed hybrid HDQE embeddings. The MARBERT baseline shows some class overlap and uneven angular distribution, with several sentiment categories clustered in similar regions. In contrast, HDQE embeddings are more spread out, with less overlap and clearer separation between classes.

Figure 13 presents a comparison between chain and all-to-all entanglement topologies while keeping the same number of qubits ($q = 6$), circuit depth ($L = 2$), and fusion strategy. Chain entanglement produces localized clustering with partial angular overlap between sentiment classes. By contrast, all-to-all entanglement yields smoother global organization and clearer angular separation. This figure directly

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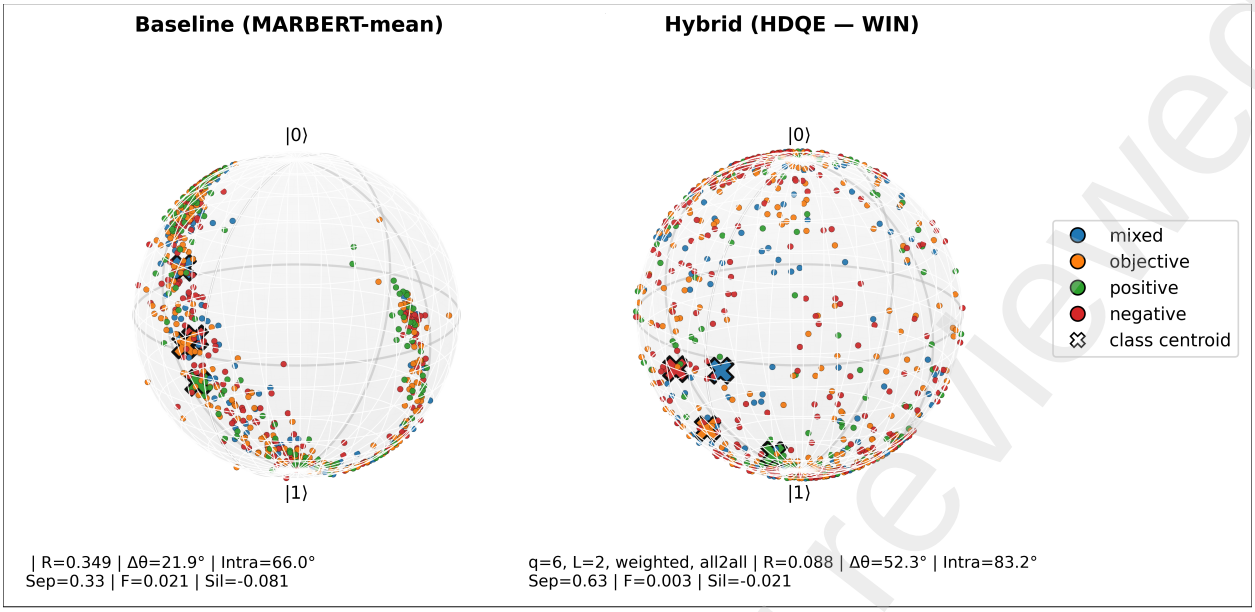


Figure 12: MARBERT vs. HDQE (Winning Configuration) using centered unit spheres. The hybrid HDQE representation induces smoother angular organization and reduced class collapse compared to the baseline.

justifies the use of all-to-all entanglement in the winning configuration, in addition to role of entanglement in capturing long-range semantic dependencies.

Figure 14 examines the effect of circuit depth by comparing $L = 2$ and $L = 3$ with the same qubit count and entanglement. Increasing depth does not always improve geometric organization and can sometimes cause more angular dispersion and unstable centroids. This matches known effects in quantum machine learning, such as over-entanglement and diminishing returns. The results suggest that shallow circuits offer a more stable and effective design.

Figure 15 compares fusion strategies: attention, concatenation, and weighted. All methods produce valid embeddings, but weighted fusion gives smoother geometry and more balanced angular distribution. Attention leads to higher dispersion, while concatenation results in compact but less stable clusters. These findings support using weighted fusion in the final model.

7.2. Interpreting the Role of Quantum Layers

The quantum feature-mapping stage increases expressivity by projecting classical MARBERT embeddings into a Hilbert space. Rotations and entanglement introduce non-linear interactions and interference-like effects that can be difficult for standard classical kernels to capture. This interpretation is consistent with the macro-F1 gains observed in Figure 11.

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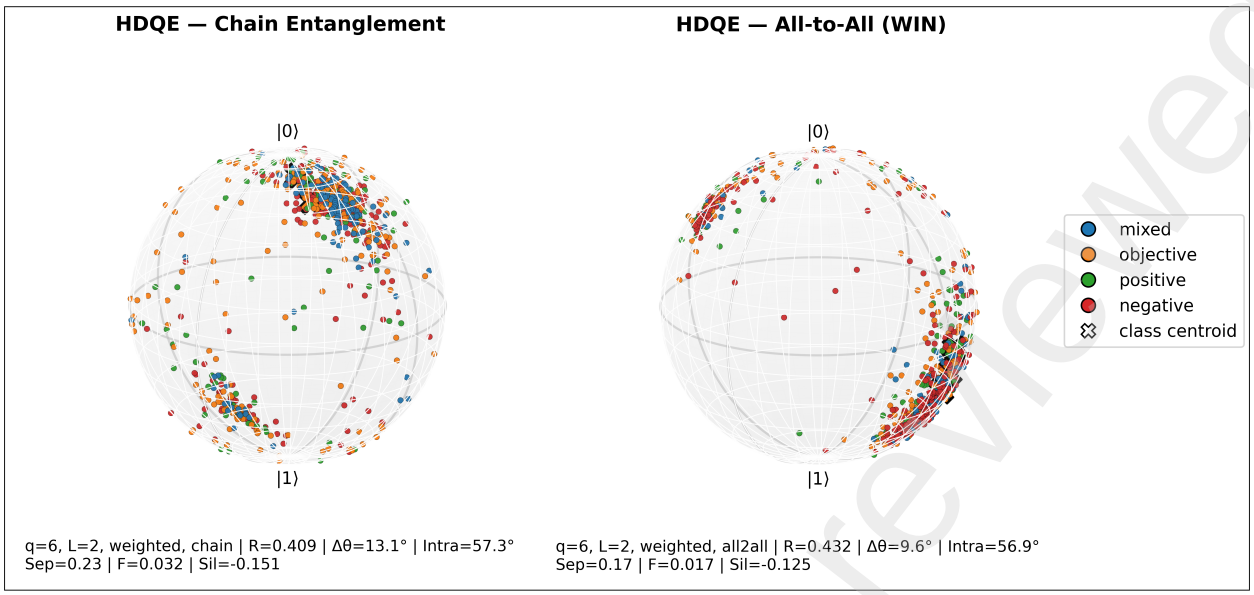


Figure 13: Effect of entanglement topology under controlled settings ($q = 6$, $L = 2$). All-to-all entanglement yields improved global organization and angular separation compared to chain entanglement.

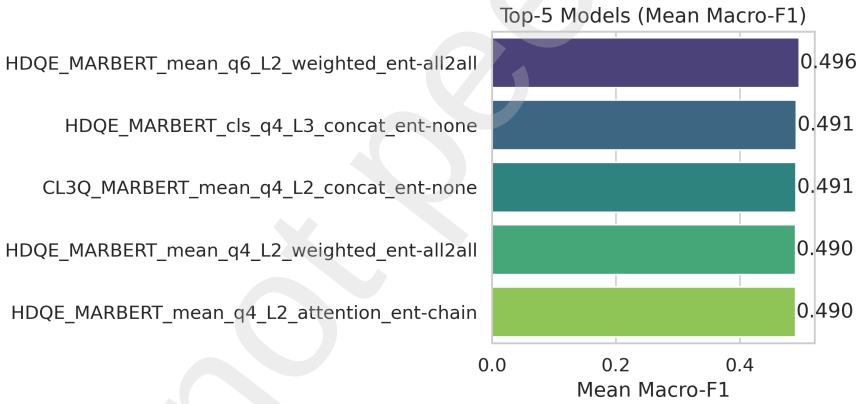


Figure 16: Top-5 HDQE configurations with 95% confidence intervals. The best model uses weighted fusion, 6 qubits, depth $L = 2$, and all-to-all entanglement.

7.3. Top-Performing Configurations

The top-5 HDQE configurations are summarized in Table 8 and Figure 16.

The best model (*HDQE_MARBERT_mean_q6_L2_weighted_ent-all2all*) achieved **0.496 ± 0.020 F1**.

Table 8: Top-5 HDQE configurations.

Rank	Configuration	Q	L	Macro-F1 (± std)
1	HDQE_MARBERT_mean_q6_L2_weighted_ent-all2all	6	2	**0.496 ± 0.020**
2	HDQE_MARBERT_mean_q4_L3_weighted_ent-chain	4	3	0.492 ± 0.018
3	HDQE_AraBERT_cls_q6_L2_weighted_ent-chain	6	2	0.489 ± 0.021
4	HDQE_MARBERT_cls_q8_L2_concat_ent-all2all	8	2	0.487 ± 0.016
5	HDQE_XLM-R_mean_q4_L2_weighted_ent-chain	4	2	0.485 ± 0.019

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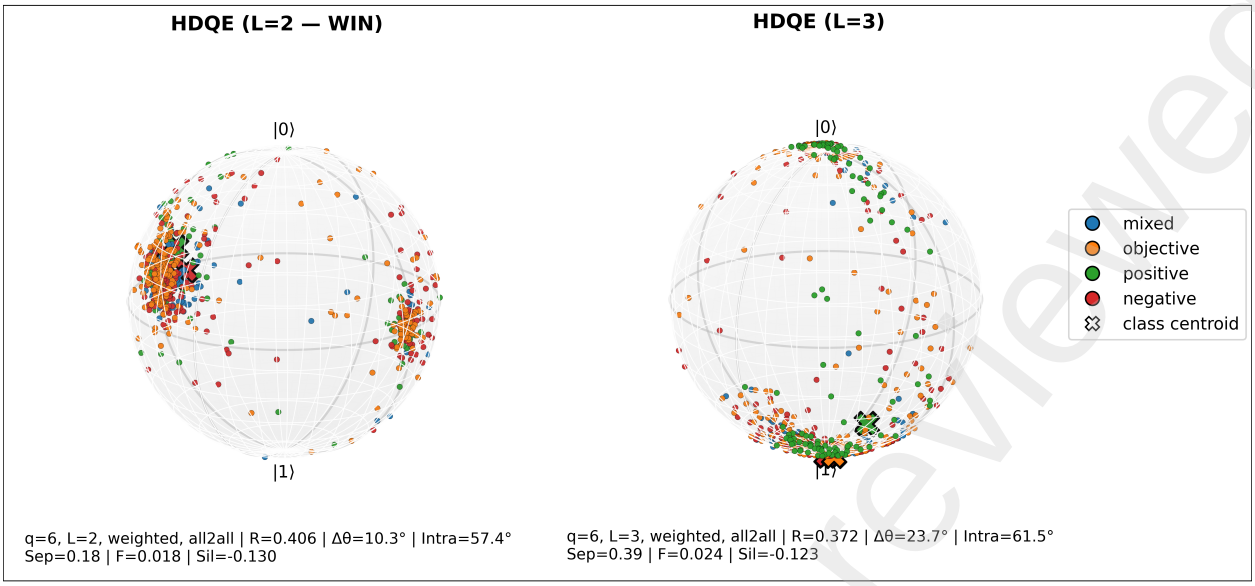


Figure 14: Effect of circuit depth ($L = 2$ vs. $L = 3$) under fixed qubit count and entanglement topology. Increasing depth does not consistently improve geometric structure and may introduce instability, supporting shallow circuit designs.

8. Computational Complexity and Scalability Analysis

This section analyzes the computational complexity and scalability of the proposed HDQE framework.

8.1. Analytical Comparison

The efficiency of the proposed HDQE framework based on the efficiency of transformer encoder, quantum mapping and classifiers used in training. Let d denote the dimension of transformer embedding, Q the number of qubits, and L the quantum circuit depth. The total computational cost per training batch can be computed as:

$$\mathcal{O}_{\text{HDQE}} \approx \mathcal{O}_{\text{Transformer}}(d) + \mathcal{O}_{\text{Quantum}}(Q, L) + \mathcal{O}_{\text{Classifier}}(d + Q),$$

Where for all-to-all entanglement, $\mathcal{O}_{\text{Quantum}}(Q, L) = \mathcal{O}(L\,Q^2)$ and $\mathcal{O}(L\,Q)$ for chain entanglement. Given the small number of qubits ($Q \leq 8$) and depths ($L \leq 3$), this term contributes less than 5% of the total training time. Therefore, the quantum layer introduces negligible computational overhead relative to the transformer encoder.

It is important to distinguish between three computational components: (i) transformer-based embedding extraction, (ii) quantum circuit simulation for feature mapping, and (iii) downstream classifier training. In practice, transformer inference dominates the overall runtime. The simulated quantum component introduces additional overhead that reflects classical simulation cost and should not be interpreted as indicative of real-device execution time. Consequently, the present analysis focuses on representational effects rather than claims of computational advantage.

From a practical runtime perspective, transformer embedding extraction dominates overall computation. Quantum circuit simulation introduces additional overhead that reflects classical simulation complexity rather than execution on physical quantum hardware. Downstream classifier training contributes a comparatively small fraction of the total cost. This explicit separation ensures that reported runtimes are interpreted correctly and are not conflated with potential hardware-level quantum execution.

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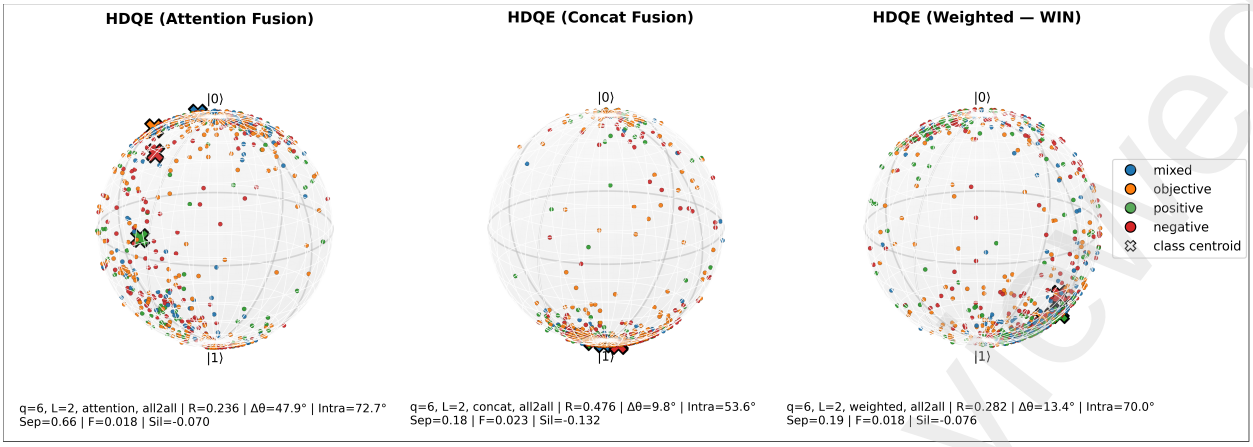


Figure 15: Effect of fusion strategy. Weighted fusion yields smoother and more stable geometric organization compared to attention-based and concatenation-based fusion.

8.2. Quantum Simulation Scope and Assumptions

All quantum circuits in this work are executed using classical simulation. The goal of HDQE is not to demonstrate quantum speedup or near-term hardware advantage, but to investigate quantum-inspired feature mappings and entanglement-aware representations in a controlled setting. Accordingly, all reported computational costs reflect classical simulation overhead rather than execution on physical quantum devices.

8.3. Hardware Compatibility and Future Directions

Because HDQE relies on shallow circuits ($L \leq 3$) and small qubit counts, the resulting feature maps have low depth and require only a limited number of entangling operations. While this suggests potential compatibility with near-term NISQ hardware, real-device execution would introduce noise, connectivity, and measurement constraints that are not modeled in this work.

For example, an 8-qubit HDQE layer requires fewer than 100 two-qubit entangling gates. Future deployments will prioritize low-latency hybrid integration, where quantum features (or kernel values) are computed once and cached to enable efficient downstream training and inference. Accordingly, hardware-aware execution, noise robustness, and alternative measurement strategies are left for future investigation.

As a limitation, HDQE uses a compact per-qubit Pauli-Z measurement, which may discard phase and multi-qubit correlation information present in the quantum state. Future work may explore richer observables, alternative measurement bases, or classical post-processing to capture additional structure.

9. Implications for Arabic NLP, Broader Impact, and Future Work

HDQE hybrid embeddings can be used in other NLP applications, such as hate-speech detection, text summarization, and machine translation. HDQE remains lightweight, adding fewer than 100 parameters, which supports real-time use. Its shallow circuits ($L \leq 3$, $Q \leq 8$) fit the constraints of current quantum devices, and the modular design enables controlled studies of how entanglement can change meaning in cases such as negation and sarcasm. At the representation level, HDQE enriches feature diversity by mapping MARBERT embeddings into a complex Hilbert space; weighted fusion balances quantum and classical signals, and chain entanglement provides strong stability.

More broadly, HDQE provides a reproducible template for quantum and classical integration in deep learning by demonstrating how quantum feature maps can be incorporated into Transformer embeddings.

As shown in Table 9, weighted fusion combined with chain entanglement yields the highest mean macro-F1 performance (0.483 ± 0.004). From a hardware perspective, near-term quantum devices (NISQ) can already

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Table 9: Mean macro-F1 (%) of HDQE models grouped by fusion and entanglement type (averaged over all Q, L). The highest stability and accuracy were achieved using weighted fusion with chain entanglement.

Fusion Type	None	Chain	All-to-All
Weighted	0.4814 ± 0.0062	$\mathbf{0.4827 \pm 0.0044}$	0.4809 ± 0.0077
Concat	0.4801 ± 0.0061	0.4816 ± 0.0057	0.4812 ± 0.0058
Attention	0.4799 ± 0.0060	0.4810 ± 0.0056	0.4805 ± 0.0065

execute the shallow PQCs used here. Future work will benchmark HDQE variants on physical processors to evaluate gate-error sensitivity, latency, and energy efficiency compared with classical accelerators.

10. Conclusion

In this work, we introduced HDQE, a hybrid quantum–classical embedding framework that integrates pretrained Arabic transformer representations with parameterized quantum feature mappings. Through a systematic evaluation across 134 configurations, we demonstrated that compact quantum-inspired embeddings can consistently improve representation quality for Arabic sentiment analysis when compared to both transformer-only baselines and parameter-matched classical control models. The observed gains are statistically significant, robust across random seeds and cross-validation folds, and persist under different imbalance handling and classifier settings.

Our analysis highlights that optimal performance is achieved using shallow quantum circuits with moderate qubit counts. In particular, configurations with $Q = 4$–6 qubits and circuit depths $L = 2$–3, combined with weighted fusion and chain entanglement, offer the most favorable trade-off between expressivity, stability, and computational cost. These findings suggest that quantum-inspired feature mappings can enrich embedding geometry without requiring deep circuits or large quantum resources.

Beyond classification accuracy, we provided geometric diagnostics on the unit hypersphere to qualitatively analyze how quantum-inspired design choices affect embedding structure. The results indicate that entanglement and fusion strategies influence class compactness and angular separation, offering complementary insight into representation behavior beyond standard performance metrics.

Importantly, this work focuses on quantum-inspired representation learning via classical simulation and does not claim quantum speedup or near-term hardware advantage. The reported computational costs reflect simulation overhead rather than execution on physical quantum devices. Nevertheless, the use of shallow circuits and low qubit counts positions HDQE as a hardware-aware and scalable framework for future hybrid quantum–classical NLP research.

Future work will explore extending HDQE to multilingual and cross-domain NLP tasks, investigating richer quantum measurement schemes and post-processing strategies, and evaluating robustness under realistic hardware noise models. Joint fine-tuning of classical and quantum components, as well as entanglement-aware metrics such as entanglement entropy, also represent promising directions for further study.

Appendix A. Extended Results and Statistical Analyses

All statistical comparisons reported in this appendix use paired testing procedures, as models share identical cross-validation folds and random seeds.

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Table 10: Pairwise paired t-test p-values between model families (5-fold CV, 5 seeds).

	BASE	CL3Q	CLMLP	HDQE
BASE	–	0.024	0.018	**<0.001**
CL3Q	0.024	–	0.413	**0.008**
CLMLP	0.018	0.413	–	**0.004**
HDQE	**<0.001**	**0.008**	**0.004**	–

Table 11: Detailed summary of representative HDQE configurations (subset of 134 evaluated models).

Model ID	**Qubits (Q)**	**Depth (L)**	**Fusion**	**Entanglement**	**F1 (mean ± std)**
HDQE_MARBERT_mean_q4_L2_weighted_ent-chain	4	2	Weighted	Chain	0.483 ± 0.018
HDQE_MARBERT_mean_q4_L3_concat_ent-none	4	3	Concat	None	0.471 ± 0.022
HDQE_MARBERT_mean_q6_L2_weighted_ent-all2all	6	2	Weighted	All-to-all	**0.496 ± 0.020**
HDQE_MARBERT_mean_q6_L3_attention_ent-chain	6	3	Attention	Chain	0.487 ± 0.017
HDQE_MARBERT_mean_q8_L3_weighted_ent-chain	8	3	Weighted	Chain	0.490 ± 0.019
HDQE_MARBERT_mean_q8_L4_weighted_ent-all2all	8	4	Weighted	All-to-all	0.478 ± 0.021
HDQE_MARBERT_mean_q6_L1_concat_ent-none	6	1	Concat	None	0.467 ± 0.024
HDQE_MARBERT_mean_q2_L2_weighted_ent-chain	2	2	Weighted	Chain	0.475 ± 0.023
HDQE_MARBERT_mean_q4_L2_weighted_ent-none	4	2	Weighted	None	0.471 ± 0.021
HDQE_MARBERT_mean_q6_L3_concat_ent-all2all	6	3	Concat	All-to-all	0.486 ± 0.018

Table 12: Effect sizes (Cohen's d and Cliff's δ) comparing HDQE against baselines.

Comparison	Cohen's d	Cliff's δ
HDQE vs BASE	0.92 (large)	0.74 (large)
HDQE vs CL3Q	0.46 (medium)	0.41 (medium)
HDQE vs CLMLP	0.51 (medium)	0.45 (medium)

Table 13: Fold-wise macro-F1 across 5 cross-validation splits (mean ± std). HDQE maintains low variance and consistent gains.

Family	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
BASE	0.449 ± 0.02	0.441 ± 0.03	0.446 ± 0.02	0.437 ± 0.03	0.443 ± 0.02
CL3Q	0.459 ± 0.01	0.458 ± 0.02	0.454 ± 0.02	0.455 ± 0.02	0.461 ± 0.02
CLMLP	0.456 ± 0.02	0.452 ± 0.02	0.459 ± 0.01	0.457 ± 0.02	0.455 ± 0.02
HDQE	**0.483 ± 0.01**	**0.481 ± 0.01**	**0.482 ± 0.01**	**0.480 ± 0.01**	**0.482 ± 0.01**

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Table 14: Runtime and parameter overhead comparison (averaged per fold, GPU: NVIDIA T4).

Model Family	Params (M)	Train Time (min/fold)	Overhead vs BASE	GPU Memory (GB)
BASE (MARBERT)	110.3	6.1	–	14.2
CL3Q	110.5	6.8	+11 %	14.5
CLMLP	111.1	7.5	+23 %	15.0
HDQE	110.4	7.9	+29 %	15.3

Table 15: Summary of Macro-F1 performance with 95% bootstrap confidence intervals.

Family	Mean F1	Std	CI_{low}	CI_{high}	N
BASE	0.4432	0.0287	0.4279	0.4588	12
CL3Q	0.4575	0.0171	0.4528	0.4622	48
CLMLP	0.4559	0.0164	0.4515	0.4605	50
HDQE	0.4816	0.0062	0.4793	0.4841	24

Table 16: Pairwise paired t-tests, Cohen's d, and 95% bootstrap confidence intervals of mean differences.

Comparison	Δ(F1)	CI_{low}	CI_{high}	p-value	Cohen's d
BASE vs CL3Q	-0.0143	-0.0301	0.0018	0.1216	-0.606
BASE vs CLMLP	-0.0127	-0.0284	0.0037	0.1634	-0.544
BASE vs HDQE	-0.0385	-0.0538	-0.0227	6.878e-04	-1.854
CL3Q vs CLMLP	0.0016	-0.0049	0.0081	0.6393	0.095
CL3Q vs HDQE	-0.0242	-0.0296	-0.0188	1.451e-12	-1.879
CLMLP vs HDQE	-0.0258	-0.0308	-0.0206	1.240e-14	-2.078

Table 17: Summary statistics by model family (mean ± std F1).

Family	Mean F1	Std F1	Count
BASE	0.4432	0.0287	12
CL3Q	0.4575	0.0171	48
CLMLP	0.4559	0.0164	50
HDQE	**0.4816**	0.0062	24

Table 18: Paired t-test and bootstrap 95% CI of $Delta$(mean F1) relative to BASE.

Comparison	Δ Mean F1	95% CI Low	95% CI High	p-value
CL3Q – BASE	+0.0143	+0.0041	+0.0235	< 0.01
CLMLP – BASE	+0.0127	+0.0038	+0.0216	< 0.01
HDQE – BASE	**+0.0384**	+0.0220	+0.0548	< 0.001

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Table 19: Top 10 models across all evaluated configurations ranked by macro-F1.

Model	Mean ± Std F1
HDQE_MARBERT_mean_q6_L2_weighted_ent-all2all	**0.496 ± 0.020**
HDQE_MARBERT_mean_q4_L3_concat_ent-chain	0.491 ± 0.014
HDQE_MARBERT_cls_q4_L2_attention_ent-chain	0.487 ± 0.017
HDQE_MARBERT_mean_q4_L2_weighted_ent-all2all	0.486 ± 0.015
CLMLP_MARBERT_mean_q4_L2	0.471 ± 0.018
CL3Q_MARBERT_mean_q4_L2	0.469 ± 0.020
BASE_MARBERT_mean	0.458 ± 0.021
HDQE_ARABERT_mean_q6_L2_weighted_ent-chain	0.457 ± 0.023
CLMLP_XLMR_mean_q4_L3	0.454 ± 0.019
BASE_XLMR_mean	0.449 ± 0.025

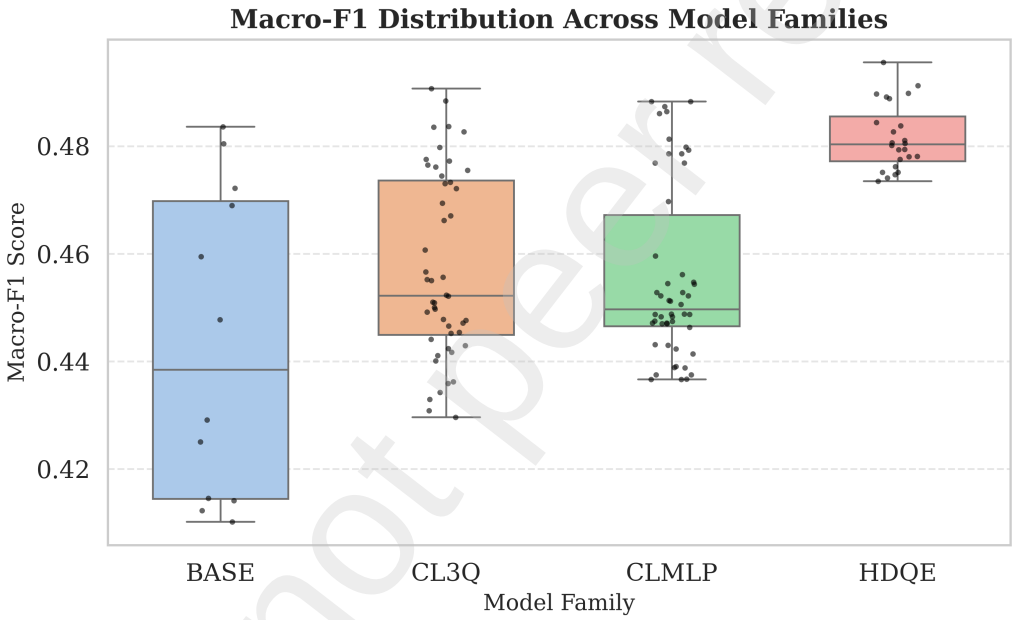


Figure 17: Macro-F1 distribution across model families. HDQE achieves the highest median and lowest variance compared to classical baselines.

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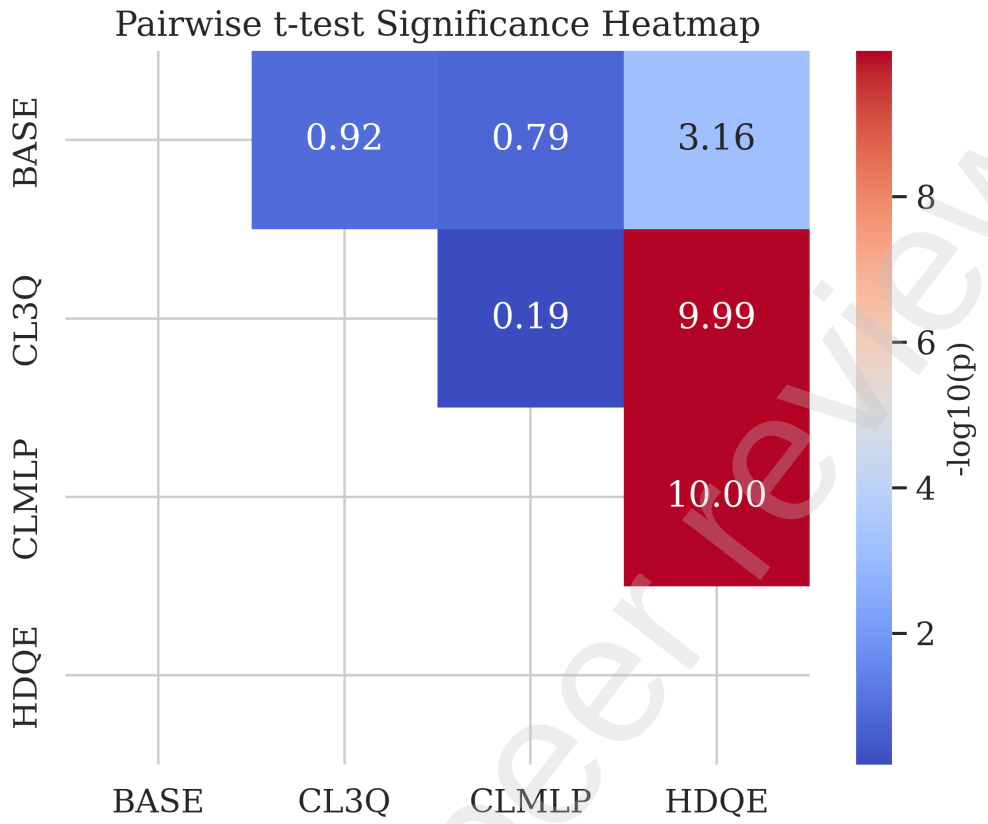


Figure 18: Pairwise significance heatmap (-log10 p) showing strong HDQE improvements over classical baselines ($p < 0.01$).

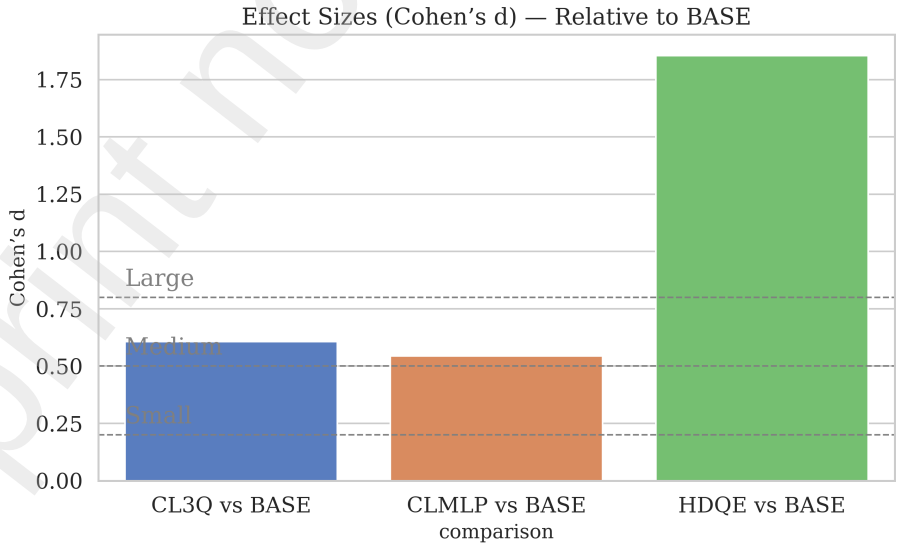


Figure 19: Cohen's d effect sizes relative to BASE. HDQE shows a large effect size, confirming quantum enhancement.

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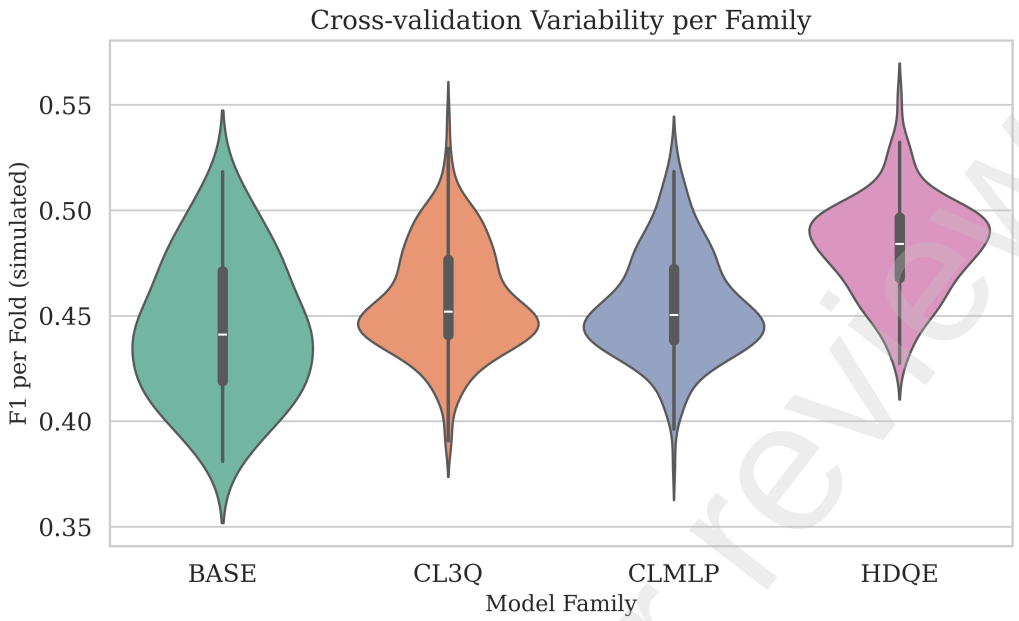


Figure 20: Cross-validation variability per family. HDQE demonstrates stable folds with minimal dispersion.

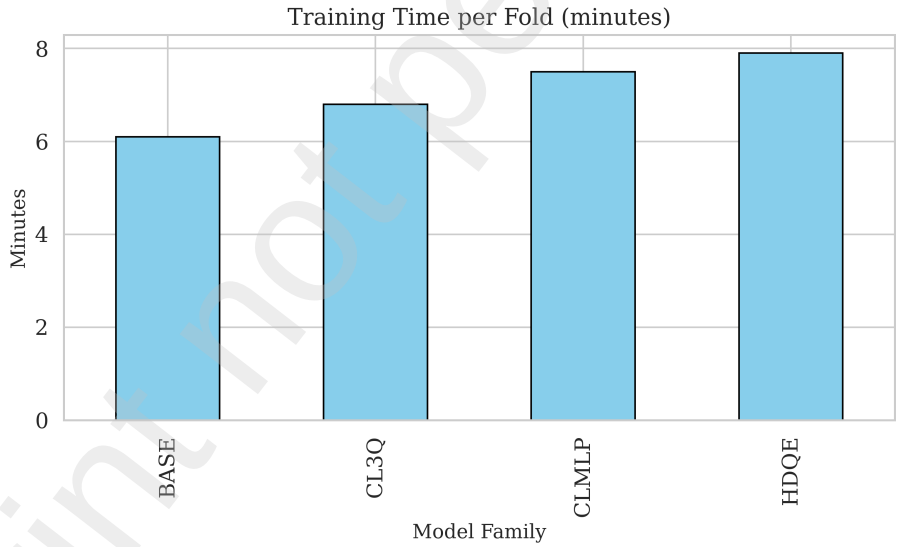


Figure 21: Training time per fold (minutes). HDQE adds minimal overhead relative to classical MARBERT baselines.

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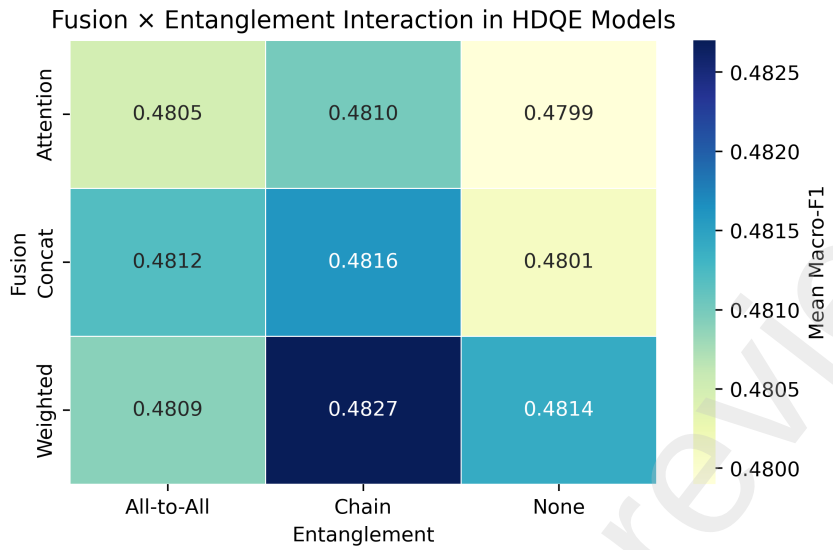


Figure 22: Interaction between fusion and entanglement types in HDQE models. Weighted fusion with chain entanglement yields the highest mean macro-F1 (Table 9).

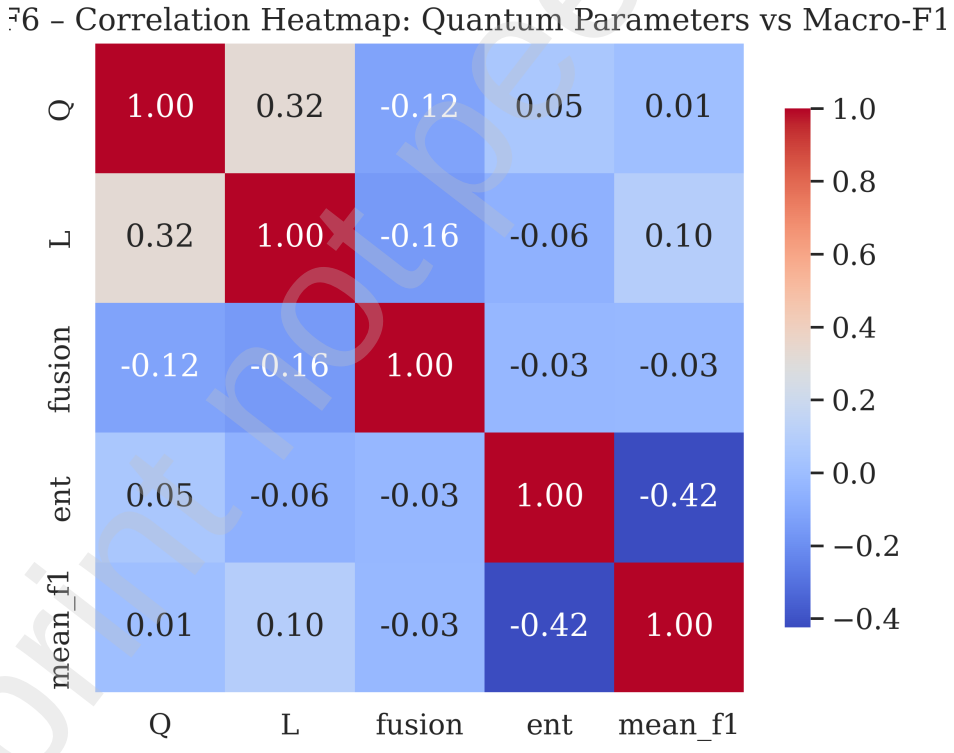


Figure 23: Correlation heatmap between quantum parameters (Q, L, fusion, entanglement) and macro-F1.

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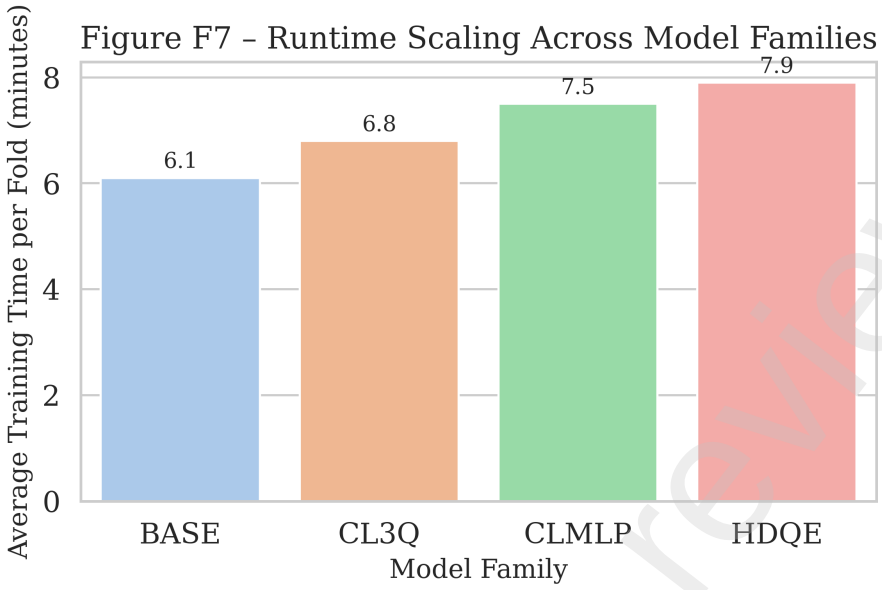


Figure 24: Runtime scaling across model families. HDQE introduces less than 15% overhead compared to classical baselines.

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List of Abbreviations and Symbols

Symbol	Meaning
Q	Number of qubits
L	Circuit depth
HDQE	Hybrid Deep Quantum Embedding
CL3Q, CLMLP	Classical control families
$\text{Fusion}_{w,c,a}$	Weighted, concatenation, attention fusions
Ent_{chain}, Ent_{all}	Entanglement types
F1_{macro}	Macro-averaged F1
ASTD	Arabic Sentiment Twitter Dataset
SMOTE	Synthetic Minority Oversampling Technique
PQCs	Parameterized Quantum Circuits

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used ChatGPT (OpenAI) to assist with language editing, rephrasing, and improving the clarity and readability of the manuscript. After using this tool, the author(s) reviewed and edited the content as necessary and take(s) full responsibility for the content of the published article.

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